ACCESS TRADES

Trades Numeracy Answer Key

Answers to every Try it set and worksheet in the Trades Numeracy Textbook, and to every exercise in the Trades Numeracy Exercise Book, with working for the longer problems. For fabrication, pipe trades, millwright and carpentry pre-apprentices.

ACCESS

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How to use this key

The key follows the books in order. For each chapter, Part 1 gives the answers to the textbook's Try it sets and Shape Canvas worksheets, and Part 2 gives the answers to the Exercise Book sets A, B and C, and to any further sets in that chapter, such as the by-hand sets and the Carpentry and the BCBC sets.

The appendices come after Chapter 13: Appendix A, Kitchen Math, from both books, and Appendix B, the Practice Assessment from the Exercise Book.

Answers that were rounded give the same number of decimal places the question asked for. Where a problem takes more than one step, the working is shown in smaller type under the answer. Your working can be different and still be right if it reaches the same answer.

1 Whole Numbers

ANSWERS

Part 1: Textbook

Try it 1.1

1. Three thousand forty-eight
2. 7,602
3. 5,000
4. 90
5. 2,450 mm (45 tens is more than 40 tens)
6. 20,050

Try it 1.2

1. 1,370
2. 1,380
3. 6,100
4. 6,000
5. 50,000
6. 800

Try it 1.4

1. 3,825
2. 1,950
3. 3,050 mm
4. 25,458

Try it 1.5

1. 3,125
2. 2,490
3. 1,125 mm
4. 1,200 mm

Try it 1.6

1. 120
2. 1,200
3. 8,000
4. 8,100 mm
5. 21 mm
6. 84

Try it 1.7

1. 12
2. 13
3. 125
4. 7 pieces (2,660 mm), offcut 340 mm
5. 4
6. 110 mm (550 ÷ 5)

Try it 1.9

1. 14:45
2. 7:20 p.m.
3. 2 h 25 min
4. 3 h 25 min
5. 7.5 hours
6. 8.2 hours
7. 3 h 45 min
8. 8 h (8 hours)

Try it 1.10

1. 1,904
2. 6,000
3. 7,368
4. 29,104
5. 246,840
6. 5,940 g

Try it 1.11

1. 14
2. 7
3. 5
4. Mean 32, median 30, mode 30
5. 6
6. Tuesday (35 rods)

Try it 1.12

1. 16
2. 6
3. 15 (13.75 rounds up to 14 spaces)
4. 400 mm
5. 6 blocks

Worksheet 1A

1. **2,050 mm**
2. **1,075 mm**
3. **4,410 mm**
4. **15 pieces, offcut 300 mm**
5. **991 mm (5,000 mm of pieces plus 9 mm of kerf)**
6. **150 mm**

Worksheet 1B

1. **600 mm**
2. **700 mm**
3. **900 + 250 + 600 + 450 + 300 + 700 = 3,200 mm**
4. **3 × 8 = 24 holes, 24 × 2 = 48 minutes**
5. **Estimate 4 × 1,200 = 4,800. Exact 4,720.**
6. **304 mm**

Worksheet 1C

1. **10 studs**
   3,600 ÷ 400 = 9 spaces. 9 + 1 = 10.
2. **400 mm o.c.**
   13 studs make 12 spaces. 4,800 ÷ 12 = 400.
3. **3,600 mm**
   7 studs make 6 spaces. 6 × 600 = 3,600.
4. **14 joists. The last space is 80 mm.**
   4,880 ÷ 400 = 12.2, round up to 13 spaces. 13 + 1 = 14.
   12 × 400 = 4,800, 4,880 − 4,800 = 80.
5. **No. The table allows 38 × 89 studs at no more than 400 mm for this wall. 16 studs.**
   6,000 ÷ 400 = 15 spaces, 15 + 1 = 16.
6. **3 pieces per board, 4 boards**
   610 + 3 = 613. 2,440 ÷ 613 = 3.98, so 3 pieces.
   10 ÷ 3 = 3.33, round up to 4 boards.
7. **Estimate 15,000 ÷ 1,200 is about 12 per row. Exact: 12 per row, 24 sheets.**

Part 2: Exercise Book

Exercise 1A

1. **Four thousand seventy-two**
2. **15,308**
3. **7,000**
4. **3,049, 3,409, 3,490, 3,940**
5. **3,460**
6. **9,000**
7. **15,000**
8. **800**
9. **4,700 (1,200 + 2,900 + 600)**
10. **1,000 (50 × 20)**
11. **5,223**
12. **3,313**
13. **4,632**
14. **5,375**
15. **480**
16. **5,160**
17. **224**
18. **15 remainder 200**
19. **65**
20. **4,632 + 2,368 = 7,000**

Exercise 1B

1. **1,525 mm**
   Total cut 4,475 mm. 6,000 − 4,475 = 1,525.
2. **61 ft. 61 ÷ 21 = 2.9, so buy 3 lengths.**
   Round up: a part length still has to be bought.
3. **112 bolts, 5 boxes**
   14 × 8 = 112. 112 ÷ 25 = 4.48, round up to 5 boxes.
4. **144 nuts**
5. **13 pieces, offcut 6 in**
   240 ÷ 18 = 13.33. 13 × 18 = 234. 240 − 234 = 6.
6. **7 in**
   48 − 3 − 3 = 42 in. 7 holes make 6 spaces. 42 ÷ 6 = 7.
7. **8 hours**
8. **Estimate 50 × 24 = 1,200. Exact 1,080 parts.**
9. **5 boxes, 34 rods left over**
   18 × 12 = 216 rods. 216 ÷ 50 = 4.32, round up to 5 boxes = 250 rods. 250 − 216 = 34.

Exercise 1C

1. **1,625 mm**
2. **4,210 mm**
3. **1,365 mm**
4. **1,080 mm**
   6 spaces × 160 = 960. 960 + 60 + 60 = 1,080.
5. **750 mm**
6. **500 mm**
7. **1,100 + 300 + 750 + 500 + 350 + 800 = 3,800 mm**
8. **4 × 6 = 24 holes, 24 × 3 = 72 minutes**
9. **75 mm shorter. The offcut is 1,550 mm.**

Exercise 1D

1. **9,351**
   Line up the ones. Column sums from the right: 21, 15, 23, 9.
2. **6,524**
   Borrow through the zeros: 10,000 = 9 thousands, 9 hundreds, 9 tens and 10 ones.
3. **3,136**
   Borrow through the zeros in the tens and hundreds.
4. **2,368**
   64 × 7 = 448, 64 × 30 = 1,920. Sum 2,368.
5. **11,638**
   506 × 3 = 1,518, 506 × 20 = 10,120. Sum 11,638.
6. **95,700**
   348 × 5 = 1,740, 348 × 70 = 24,360, 348 × 200 = 69,600. Sum 95,700.
7. **287,854**
   709 × 6 = 4,254, 709 × 400 = 283,600. Sum 287,854.
8. **1,214**
   Start with 7. 6 goes into 7 once: 1 × 6 = 6, and 7 − 6 = 1. Bring down the 2 to make 12. 6 goes into 12 twice: 2 × 6 = 12, and 12 − 12 = 0. Bring down the 8 to make 8. 6 goes into 8 once: 1 × 6 = 6, and 8 − 6 = 2. Bring down the 4 to make 24. 6 goes into 24 4 times: 4 × 6 = 24, and 24 − 24 = 0. Check: 1,214 × 6 = 7,284.
9. **1,141 r 7**
   Start with 9. 8 goes into 9 once: 1 × 8 = 8, and 9 − 8 = 1. Bring down the 1 to make 11. 8 goes into 11 once: 1 × 8 = 8, and 11 − 8 = 3. Bring down the 3 to make 33. 8 goes into 33 4 times: 4 × 8 = 32, and 33 − 32 = 1. Bring down the 5 to make 15. 8 goes into 15 once: 1 × 8 = 8, and 15 − 8 = 7. Check: 1,141 × 8 + 7 = 9,135.
10. **252**
    Start with 60. 24 goes into 60 twice: 2 × 24 = 48, and 60 − 48 = 12. Bring down the 4 to make 124. 24 goes into 124 5 times: 5 × 24 = 120, and 124 − 120 = 4. Bring down the 8 to make 48. 24 goes into 48 twice: 2 × 24 = 48, and 48 − 48 = 0. Check: 252 × 24 = 6,048.
11. **265 r 20**
    Start with 85. 32 goes into 85 twice: 2 × 32 = 64, and 85 − 64 = 21. Bring down the 0 to make 210. 32 goes into 210 6 times: 6 × 32 = 192, and 210 − 192 = 18. Bring down the 0 to make 180. 32 goes into 180 5 times: 5 × 32 = 160, and 180 − 160 = 20. Check: 265 × 32 + 20 = 8,500.
12. **301 r 35**
    Start with 135. 45 goes into 135 3 times: 3 × 45 = 135, and 135 − 135 = 0. Bring down the 8 to make 8. 45 goes into 8 zero times. Write 0. Bring down the 0 to make 80. 45 goes into 80 once: 1 × 45 = 45, and 80 − 45 = 35. Check: 301 × 45 + 35 = 13,580.

Exercise 1K

1. **19 studs. At 600 mm, 13 studs, which is 6 fewer.**
   7,200 ÷ 400 = 18, 18 + 1 = 19. 7,200 ÷ 600 = 12, 12 + 1 = 13.
2. **600 mm, and 3.0 m**
3. **16 studs**
   4,500 ÷ 300 = 15 spaces, 15 + 1 = 16.
4. **11 studs, last space 300 mm**
   3,900 ÷ 400 = 9.75, round up to 10 spaces. 9 × 400 = 3,600. 3,900 − 3,600 = 300.
5. **14 joists**
   5,000 ÷ 400 = 12.5, round up to 13 spaces. 13 + 1 = 14.
6. **5 pieces per board, 4 boards**
   457 + 3 = 460. 2,438 ÷ 460 = 5.3, so 5. 18 ÷ 5 = 3.6, round up to 4.
7. **21,600 mm, 5 lengths**
   3 × 7,200 = 21,600. 21,600 ÷ 4,880 = 4.43, round up to 5.
8. **12 sheets**
   7,200 ÷ 1,220 = 5.9, round up to 6 per row. 2 × 6 = 12.
9. **8 hours each, 32 hours for the crew**
   7:15 a.m. to 3:45 p.m. is 8 h 30 min. 8 h 30 min − 30 min = 8 h. 4 × 8 = 32.
10. **Mean 14, median 13, mode 13**
    Sum 70, 70 ÷ 5. In order: 9, 13, 13, 16, 19.

2 Your Calculator: TI-30Xa and Sharp EL-531XT

ANSWERS

Part 1: Textbook

Try it 2.4

1. 81, √x (answer 9)
2. √, 81, = (answer 9)
3. 225
4. 14
5. 64
6. 4

Try it 2.5

1. 7
2. −24
3. −32
4. −4

Try it 2.6

1. 160
2. 81
3. 1,000
4. 628.32
5. 5,026.55
6. 3

Try it 2.7

1. $4\frac{1}{4}$ (4.25)
2. 0.625
3. 2.375
4. 90

Try it 2.10

1. 0.5
2. 0.7071
3. 0.3640
4. 30°
5. 36.87°
6. 191.51

Try it 2.12

1. $75\frac{1}{2}''$
2. $127\frac{3}{4}''$
3. $33\frac{1}{16}''$
4. $10' 1\frac{3}{4}''$ ($121\frac{3}{4}''$)
5. $8' 3\frac{3}{4}''$ ($12 \times 12 - (3 \times 12 + 8\frac{1}{4}) = 99\frac{3}{4}''$)
6. 2,353 mm

Worksheet 2A

1. 2,845. TI-30Xa: 2450, +, 1375, −, 980, =. Sharp: 2450, +, 1375, −, 980, =.
2. 324. TI-30Xa: 18, x^2. Sharp: 18, x^2, =.
3. 23. TI-30Xa: 529, √x. Sharp: √, 529, =.
4. 12. TI-30Xa: 12, +⇄−, ×, 4, +, 60, =. Sharp: +/−, 12, ×, 4, +, 60, =.
5. D = 120 mm. π × 120 = 376.99 mm. TI-30Xa: π, ×, 120, =. Sharp: π, ×, 120, =.
6. $4\frac{3}{8}$ in. The TI-30Xa shows 4⌟3⌟8 and the Sharp shows 4┌3┌8. TI-30Xa: 2, a b/c, 3, a b/c, 4, +, 1, a b/c, 5, a b/c, 8, =. Sharp: 2, a b/c, 3, a b/c, 4, +, 1, a b/c, 5, a b/c, 8, =.
7. 68 kg. TI-30Xa: 340, ×, 20, 2nd, %, =. Sharp: 340, ×, 20, 2ndF, %.
8. 338.09. TI-30Xa: 25, SIN, ×, 800, =. Sharp: sin, 25, ×, 800, =.

Worksheet 2B

1. A: a b/c, the fraction key. B: +⇄−, change sign (after the digits). C: DRG, changes the angle unit. D: STO, stores the display in memory. E: 2ndF, for the orange second functions. F: +/−, change sign (before the number). G: √, square root (before the number). H: DEL, deletes a character.
2. $3\frac{5}{16}$
3. The square root of 225. The answer is 15.
4. Press DRG until DEG shows (RAD, then GRAD, then DEG: two presses).

Part 2: Exercise Book

Exercise 2A

1. **4,147. TI-30Xa: 3640, +, 1295, −, 788, =. Sharp: 3640, +, 1295, −, 788, =.**
2. **529. TI-30Xa: 23, x^2. Sharp: 23, x^2, =.**
3. **37. TI-30Xa: 1369, √x. Sharp: √, 1369, =.**
4. **343. TI-30Xa: 7, 2nd, x^3. Sharp: 7, x^3, =.**
5. **7. TI-30Xa: 343, 2nd, $\sqrt[3]{x}$. Sharp: 2ndF, $\sqrt[3]{}$, 343, =.**
6. **−50. TI-30Xa: 15, +⇄−, ×, 6, +, 40, =. Sharp: +/−, 15, ×, 6, +, 40, =.**
7. **243. TI-30Xa: 3, y^x, 5, =. Sharp: 3, y^x, 5, =.**
8. **282.74. TI-30Xa: π, ×, 90, =. Sharp: π, ×, 90, =.**
9. **$3\frac{13}{16}$ (TI-30Xa 3⌟13⌟16, Sharp 3 ┌13 ┌16). TI-30Xa: 2, a b/c, 3, a b/c, 8, +, 1, a b/c, 7, a b/c, 16, =. Sharp: 2, a b/c, 3, a b/c, 8, +, 1, a b/c, 7, a b/c, 16, =.**
10. **0.875. TI-30Xa: 7, a b/c, 8, 2nd, F⇄D. Sharp: 7, a b/c, 8, =, a b/c.**
11. **81. TI-30Xa: 450, ×, 18, 2nd, %, =. Sharp: 450, ×, 18, 2ndF, %.**
12. **1,350. TI-30Xa: 1250, +, 8, 2nd, %, =. Sharp: 1250, +, 8, 2ndF, %.**
13. **0.7880. TI-30Xa: 52, SIN. Sharp: sin, 52, =.**
14. **30.96°. TI-30Xa: 0.6, 2nd, TAN^{-1}. Sharp: 2ndF, $\tan^{-1}$, 0.6, =.**
15. **1,375. TI-30Xa: 640, +, 385, =, STO, 1, ON/C, 2400, −, RCL, 1, =. Sharp: 640, +, 385, STO, A, ON/C, 2400, −, RCL, A, =.**
16. **0.125. TI-30Xa: 8, 1/x. Sharp: 8, 2ndF, x^{-1}, =.**
17. **−16. Use the change-sign key, not the minus key. TI-30Xa: 25, +⇄−, +, 9, =. Sharp: +/−, 25, +, 9, =.**
18. **1,369 mm^2. TI-30Xa: 37, x^2. Sharp: 37, x^2, =.**

Exercise 2B

1. **359.1 mm**
   TI-30Xa: π, ×, 114.3, =. Sharp: π, ×, 114.3, =.
2. **1,130,973 mm^2**
   TI-30Xa: 600, x^2, ×, π, =. Sharp: 600, x^2, ×, π, =.
3. **$28\frac{1}{8}$ in**
   TI-30Xa: 72, −, 3, ×, 14, a b/c, 5, a b/c, 8, =. Sharp: 72, −, 3, ×, 14, a b/c, 5, a b/c, 8, =.
4. **425.6 m**
   TI-30Xa: 380, +, 12, 2nd, %, =. Sharp: 380, +, 12, 2ndF, %.
5. **75,600 mm**
   One frame: 4,200 mm. TI-30Xa: 1240, +, 860, +, 1240, +, 860, =, STO, 1, ON/C, RCL, 1, ×, 18, =. Sharp: 1240, +, 860, +, 1240, +, 860, STO, A, ON/C, RCL, A, ×, 18, =.
6. **1,500 mm**
   TI-30Xa: (, 1200, x^2, +, 900, x^2,), √x. Sharp: √, (, 1200, x^2, +, 900, x^2,), =.
7. **10.6°**
   TI-30Xa: (, 450, ÷, 2400,), 2nd, TAN^{-1}. Sharp: 2ndF, $\tan^{-1}$, (, 450, ÷, 2400,), =.
8. **−15 (15 below the setting)**
   TI-30Xa: 11, +⇄−, −, 4, =. Sharp: +/−, 11, −, 4, =.

Exercise 2C

1. **A: π. B: x^2, squares a number. C: RCL, recalls a memory. D: 2nd, for the second functions printed above the keys. E: y^x, powers. F: a b/c, the fraction key. G: RCL, recalls a memory. H: HOME, returns to normal mode.**
2. **$6\frac{11}{16}$**
3. **18,240**
4. **$3^4 = 81$**
5. **20,000,000,000 (2×10^{10})**
6. **One press (the order is DEG, RAD, GRAD, then back to DEG).**
7. **D = 100 mm. π × 100 = 314.16 mm. Doubled: π × 200 = 628.32 mm, twice as far.**

Exercise 2K

1. **$112\frac{3}{4}$″. TI-30Xa: 9, ×, 12, +, 4, a b/c, 3, a b/c, 4, =. Sharp: 9, ×, 12, +, 4, a b/c, 3, a b/c, 4, =.**
2. **$179\frac{5}{8}$″. TI-30Xa: 14, ×, 12, +, 11, a b/c, 5, a b/c, 8, =. Sharp: 14, ×, 12, +, 11, a b/c, 5, a b/c, 8, =.**
3. **9′ $5\frac{1}{8}$″ ($113\frac{1}{8}$″). TI-30Xa: 5, ×, 12, +, 6, a b/c, 3, a b/c, 16, +, 3, ×, 12, +, 10, a b/c, 15, a b/c, 16, =. Sharp: 5, ×, 12, +, 6, a b/c, 3, a b/c, 16, +, 3, ×, 12, +, 10, a b/c, 15, a b/c, 16, =.**
4. **8′ $2\frac{5}{8}$″ ($98\frac{5}{8}$″). TI-30Xa: 16, ×, 12, −, (, 7, ×, 12, +, 9, a b/c, 3, a b/c, 8,), =. Sharp: 16, ×, 12, −, (, 7, ×, 12, +, 9, a b/c, 3, a b/c, 8,), =.**
5. **$41\frac{1}{2}$″ (3′ $5\frac{1}{2}$″). TI-30Xa: (, 10, ×, 12, +, 4, a b/c, 1, a b/c, 2,), ÷, 3, =. Sharp: (, 10, ×, 12, +, 4, a b/c, 1, a b/c, 2,), ÷, 3, =.**
6. **2,467 mm**

   $97\frac{1}{8}$″ × 25.4 = 2,466.975 mm. TI-30Xa: 8, ×, 12, +, 1, a b/c, 1, a b/c, 8, =, ×, 25.4, =. Sharp: (, 8, ×, 12, +, 1, a b/c, 1, a b/c, 8,), ×, 25.4, =.
7. **45′ 0$\frac{1}{2}$″ ($540\frac{1}{2}$″). TI-30Xa: 2, ×, (, 12, ×, 12, +, 6, a b/c, 1, a b/c, 2, +, 9, ×, 12, +, 11, a b/c, 3, a b/c, 4,), =. Sharp: 2, ×, (, 12, ×, 12, +, 6, a b/c, 1, a b/c, 2, +, 9, ×, 12, +, 11, a b/c, 3, a b/c, 4,), =.**
8. **9′ 0$\frac{3}{4}$″ ($108\frac{3}{4}$″). TI-30Xa: 15, ×, 7, a b/c, 1, a b/c, 4, =. Sharp: 15, ×, 7, a b/c, 1, a b/c, 4, =.**
9. **12′ 3″ (147″). TI-30Xa: 14, ×, 10, a b/c, 1, a b/c, 2, =. Sharp: 14, ×, 10, a b/c, 1, a b/c, 2, =.**
10. **$10\frac{1}{16}$″**

    255 ÷ 25.4 × 16 = 160.63 sixteenths, rounds to 161 = $10\frac{1}{16}$″. Rounding up keeps the run above the minimum.
11. **12′ $1\frac{5}{8}$″ ($145\frac{5}{8}$″)**

    3,700 ÷ 25.4 × 16 = 2,330.71 sixteenths, round down to 2,330 = $145\frac{5}{8}$″ = 12′ $1\frac{5}{8}$″.
12. **8′ $5\frac{3}{8}$″ ($101\frac{3}{8}$″ = 8 × 12 + $5\frac{3}{8}$)**
13. **There are 12 inches in a foot, not 10. 8 × 12 + 6 = 102 in, and 102 ÷ 12 = 8.5 ft.**

3 Order of Operations

ANSWERS

Part 1: Textbook

Try it 3.2

1. 18
2. 17
3. 18
4. 20

Try it 3.3

1. 10
2. 5
3. 45
4. 4,000

Try it 3.4

1. 16
2. 18
3. 36
4. 18

Try it 3.5

1. 24
2. 6
3. 32
4. 25

Try it 3.6

1. 9
2. 306
3. 20
4. 400

Try it 3.8

1. −12 °C (it is further left on the number line)
2. 5
3. −5
4. −11
5. 11
6. −42
7. 8
8. 25
9. 25 mm (−10 − (−35))
10. 0 mm

Try it 3.9

1. 10
2. 11
3. 13 (16 × 12 ÷ 16 + 1)
4. Round 4,100 ÷ 400 = 10.25 up to 11 spaces, then 12 studs.
5. 2,800 mm
6. 3,825 mm
7. 190 mm. Yes, it is between 125 mm and 200 mm.
8. 3,800 mm. No, it is more than 3.7 m (3,700 mm).

Worksheet 3A

1. 135,000 + 225,000 = 360,000 mm²
2. 2 × (600 + 250) = 1,700 mm. With L = 1,200 mm: 2 × (1,200 + 250) = 2,900 mm.
3. π × (1,600 − 625) = π × 975 = 3,063.05 mm²
4. (1,350 − 150) ÷ 8 = 150 mm

Worksheet 3B

1. 16
2. 16
3. 18 and 36. In 2 × 3² the square belongs to the 3 only.
4. 32 (16 ÷ 4 = 4, 4 × 2 = 8, 40 − 8 = 32)
5. (12 + 8) ÷ 4 = 5
6. 3 × (5 − 2) + 4 = 13
7. 10 and 14
8. (45 + 27) ÷ (14 − 6) = gives 9
9. 6,000 − (4,500 + 9) = 1,491 mm
10. 5 × 4² = 80. It grew by 35 (from 45 to 80).
11. The calculator divided 60 ÷ 3 first. Key (120 + 60) ÷ 3 = to get 60 m per job.

Part 2: Exercise Book

Exercise 3A

1. 33
2. 44
3. 26
4. 32
5. 4
6. 3
7. 36
8. 16
9. 20
10. 48
11. 10
12. 16
13. 4
14. 25
15. 30
16. 29
17. 12
18. 2.5
19. (4 + 6) × 3 = 30
20. 24 ÷ (6 − 2) = 6

Exercise 3B

1. 6 × 8 + 4 = 52 bolts
2. 4,450 mm
   Brackets first: 2,225. Then 2 × 2,225.
3. 6 in
   4 × 27 = 108, 2 × 15 = 30, 108 + 30 = 138, 144 − 138 = 6.
4. 3,111.67 mm
5. 13 in
6. 2,840 mm
7. Right: 3 × 8 + 2 × 10 = 44 hours. The wrong one gives 260.
8. The calculator divided 300 ÷ 4 first. Key (1500 − 300) ÷ 4 = to get 300 mm.

Exercise 3C

1. 140,000 + 300,000 = 440,000 mm²
2. L = 1,500 − 600 = 900 mm
3. 1,188 mm
4. π × (3,600 − 2,025) = 4,948.01 mm². With r = 30 mm: π × (3,600 − 900) = 8,482.30 mm².
5. 230 mm

Exercise 3D

1. −11, −7, −2, 0, 4, 9
2. −6
3. −15
4. 19
5. −54
6. 56
7. −8
8. 16 and −16. In -4^2 the square belongs to the 4 only.
9. 4 °C at 14:00 and −4 °C at 20:00
10. 1,400 mm
    −450 − (−1,850) = −450 + 1,850
11. Differences +2, −2, −1, −3, +1, −2. Total −5 mm. Average −0.8 mm.
    Average: (+2 −2 −1 −3 +1 −2) ÷ 6
12. −2 degrees per hour
    (−12 − 6) ÷ 9 = (−18) ÷ 9
13. −10 °C

Exercise 3K

1. 19
   7,200 ÷ 400 + 1
2. 6
   3,000 ÷ 600 + 1
3. 16
   20 × 12 ÷ 16 + 1
4. 15
   5,300 ÷ 400 = 13.25, round up to 14 spaces, then + 1
5. 9 studs
   (14 × 12 + 6) ÷ 24 = 7.25, round up to 8 spaces, + 1
6. No, 600 mm is more than 400 mm. At 400 mm o.c.: 13 studs.
7. 6,000 ÷ 400 + 1 = 16 studs. 6,000 ÷ (400 + 1) = 14.96 is not a stud count.
8. 4,000 mm (11 studs)
9. 3,240 mm
10. 2,405 mm
11. 186 mm. Yes, it is between 125 mm and 200 mm.
12. 14 risers: 211.43 mm, over 200 mm. 15 risers: 197.33 mm, within the limit. Use 15 risers.
13. 3,760 mm. No, it is more than 3,700 mm.

4 Fractions

ANSWERS

Part 1: Textbook

Try it 4.1

1. $\frac{3}{4}$
2. 7
3. 16
4. $\frac{5}{12}$ used, $\frac{7}{12}$ left
5. $\frac{1}{8}$ in
6. No. The numerator 9 is bigger than the denominator 8, so it is worth more than one whole.

Try it 4.2

1. $\frac{7}{16}$ in
2. $1\frac{1}{4}$ in
3. $1\frac{15}{16}$ in
4. $2\frac{5}{8}$ in
5. $3\frac{3}{16}$ in
6. $3\frac{7}{8}$ in

Try it 4.3

1. $\frac{5}{8}$
2. $\frac{7}{8}$
3. $\frac{5}{8}$
4. $\frac{3}{4}$
5. $\frac{10}{16}$
6. $\frac{24}{32}$
7. $\frac{7}{16}$
8. $\frac{5}{32}$
9. $\frac{5}{8}$, $\frac{11}{16}$, $\frac{3}{4}$
10. $\frac{2}{3}$ ($\frac{16}{24}$ is bigger than $\frac{15}{24}$)

Try it 4.4

1. $1\frac{3}{4}$
2. $1\frac{13}{16}$
3. $2\frac{5}{8}$
4. $\frac{19}{8}$
5. $\frac{11}{2}$
6. $\frac{31}{16}$

Try it 4.5

1. 2 a b/c 7 a b/c 16
2. $1\frac{3}{8}$
3. $\frac{15}{16}$
4. $\frac{7}{8}$
5. $\frac{19}{4}$
6. $\frac{5}{8}$

Try it 4.6

1. $\frac{13}{16}$
2. $1\frac{3}{8}$
3. $17\frac{5}{8}$
4. $52\frac{13}{16}$ in
5. $\frac{9}{16}$
6. $3\frac{5}{8}$
7. $6\frac{9}{16}$
8. $13\frac{5}{16}$ in

Try it 4.7

1. $1\frac{1}{2}$
2. 12
3. $3\frac{3}{8}$
4. $14\frac{1}{4}$ in
5. $\frac{8}{5}$
6. $\frac{7}{16}$
7. 6
8. $9\frac{3}{4}$ in

Try it 4.8

1. $23\frac{11}{16}$ in
2. 5 pieces
3. $46\frac{5}{16}$ in from either end
4. 46 mm
5. 38 mm

Worksheet 4A

1. A: $\frac{13}{16}$ in, B: $1\frac{3}{8}$ in, C: $2\frac{15}{16}$ in
2. D: $\frac{5}{32}$ in, E: $\frac{11}{32}$ in
3. F: $\frac{11}{64}$ in, G: $\frac{29}{64}$ in
4. $1\frac{3}{8}$ in is 6 sixteenths past the 1 in line. $2\frac{11}{16}$ in is 11 sixteenths past the 2 in line.
5. $21\frac{3}{8}$ in

   $96 - (24\frac{3}{8} + 31\frac{1}{2} + 18\frac{3}{4}) = 96 - 74\frac{5}{8} = 21\frac{3}{8}$
6. $11\frac{15}{16}$ in

   $40 - 12\frac{5}{8} - 15\frac{7}{16} = 11\frac{15}{16}$
7. $21\frac{1}{2}$ in

   $144 - 2 \times 36\frac{3}{8} - 2 \times 24\frac{5}{8} - 4 \times \frac{1}{8} = 21\frac{1}{2}$
8. 43 in in total. $\frac{1}{2}$ in more, for a new total of $43\frac{1}{2}$ in.
9. $14\frac{3}{8}$ in

   $72 \div 5 = 14\frac{2}{5}$. $\frac{2}{5} \times 16 = 6.4$, nearest 6 sixteenths.

Worksheet 4B

1. $7\frac{1}{4}$ in

   $10\frac{1}{2} - 3\frac{1}{4} = 7\frac{1}{4}$
2. $8\frac{3}{4}$ in

   $6\frac{1}{8} + 2\frac{5}{8} = 8\frac{3}{4}$
3. $1\frac{3}{16}$ in

   $\frac{6}{16} + \frac{8}{16} + \frac{5}{16} = \frac{19}{16} = 1\frac{3}{16}$
4. 12 pieces, offcut $1\frac{1}{2}$ in

   Each piece uses $2\frac{1}{4} + \frac{1}{8} = 2\frac{3}{8}$ in. $30 \div 2\frac{3}{8} = 12\frac{12}{19}$, so 12 pieces. Offcut $30 - 12 \times 2\frac{3}{8} = 1\frac{1}{2}$
5. $2\frac{3}{8}$ in

   $10\frac{1}{2} - 3\frac{3}{8} - 4\frac{3}{4} = 2\frac{3}{8}$
6. $12\frac{1}{4}$ in
7. $5\frac{1}{4}$ in

   $(36 - 2 \times 2\frac{1}{4}) \div 6 = \frac{63}{2} \div 6 = 5\frac{1}{4}$

Worksheet 4C

1. D = $46\frac{5}{16}$ in, E = $47\frac{7}{8}$ in, F = $48\frac{3}{16}$ in
2. $14\frac{5}{16}$ in

   $96 - (28\frac{5}{8} + 33\frac{1}{4} + 19\frac{7}{16}) - 3 \times \frac{1}{8} = 14\frac{5}{16}$
3. 8 blocks, offcut 3 in

   Each block uses $14\frac{5}{8}$ in. $120 \div 14\frac{5}{8} = 8\frac{8}{39}$
4. $39\frac{3}{16}$ in
5. $71\frac{1}{2}$ mm (71.5 mm)
6. $61\frac{1}{3}$ mm deep, within 92 mm of the bearing edge
7. $93\frac{1}{3}$ mm must stay, notch up to $46\frac{2}{3}$ mm
8. 41 mm

Part 2: Exercise Book

Exercise 4A

1. $\frac{5}{12}$. Yes, the numerator is smaller than the denominator.
2. $\frac{19}{8} = 2\frac{3}{8}$
3. A: $\frac{9}{16}$ in, B: $1\frac{5}{8}$ in, C: $2\frac{1}{4}$ in, D: $2\frac{13}{16}$ in
4. E: $\frac{7}{32}$ in, F: $\frac{15}{32}$ in
5. G: $\frac{37}{64}$ in, H: $\frac{53}{64}$ in

   G is 5 small spaces past the $\frac{1}{2}$ in line ($\frac{32}{64}$). H is 5 small spaces past the $\frac{3}{4}$ in line ($\frac{48}{64}$).
6. $\frac{7}{16}$ and $\frac{9}{16}$
7. $\frac{28}{32}$ and $\frac{20}{64}$
8. $\frac{5}{8}$ in, by $\frac{1}{16}$ in
9. $\frac{3}{8}$, $\frac{13}{32}$, $\frac{7}{16}$
10. $4\frac{5}{8}$ and $3\frac{5}{16}$
11. $\frac{71}{16}$ and $\frac{27}{4}$
12. $3\frac{9}{16}$. The TI-30Xa shows 57␣16, which is $\frac{57}{16}$.
13. $\frac{2}{5}$

    $2\frac{1}{2} = \frac{5}{2}$, turned upside down is $\frac{2}{5}$
14. (a) $1\frac{1}{16}$ (b) $10\frac{5}{16}$
15. (a) $3\frac{5}{16}$ (b) $4\frac{3}{8}$
16. (a) $7\frac{1}{2}$ (b) $4\frac{3}{8}$
17. (a) 4 (b) $5\frac{1}{4}$

Exercise 4B

1. $15\frac{5}{16}$ in

   $96 - (27\frac{5}{8} + 19\frac{3}{16} + 33\frac{1}{2}) - 3 \times \frac{1}{8} = 96 - 80\frac{5}{16} - \frac{3}{8}$

   $= 15\frac{5}{16}$
2. $41\frac{3}{16}$ in
3. $1\frac{1}{2}$ in
4. 6 shims

   $\frac{3}{16} \div \frac{1}{32} = \frac{3}{16} \times \frac{32}{1} = 6$
5. $7\frac{13}{16}$ in

   $15\frac{5}{8} \div 2 = \frac{125}{8} \times \frac{1}{2} = \frac{125}{16}$
6. $5\frac{9}{16}$ in

   $(48 - 2 \times 1\frac{3}{4}) \div 8 = 44\frac{1}{2} \div 8 = 5\frac{9}{16}$
7. 15 spacers, offcut $2\frac{1}{2}$ in

   Each spacer uses $4\frac{3}{8} + \frac{1}{8} = 4\frac{1}{2}$ in. $70 \div 4\frac{1}{2} = 15\frac{5}{9}$, so 15 spacers. $15 \times 4\frac{1}{2} = 67\frac{1}{2}$ in, offcut $2\frac{1}{2}$ in.
8. 2 in

   $2\frac{3}{8} - 2 \times \frac{3}{16} = 2\frac{3}{8} - \frac{3}{8}$
9. No. The key is $\frac{1}{16}$ in too wide.
10. 1,800 mm
11. 30 in and 15 in
12. $16\frac{11}{16}$ in

    $50 \div 3 = 16\frac{2}{3}$. $\frac{2}{3} \times 16 = 10.67$, nearest 11 sixteenths.

Exercise 4C

1. **$8\frac{7}{8}$ in**

   $12\frac{1}{4} - 3\frac{3}{8}$

2. **$6\frac{5}{8}$ in**

   $9\frac{1}{2} - 2\frac{7}{8}$

3. **$4\frac{1}{4}$ in**

   $14\frac{1}{2} - 2\frac{5}{8} - 3\frac{1}{4} - 4\frac{3}{8} = 4\frac{1}{4}$

4. **$1\frac{7}{8}$ in**

   $(30 - 5 \times 5\frac{1}{4}) \div 2$

5. **$33\frac{1}{2}$ in, then $33\frac{3}{4}$ in**

   $4 \times 8\frac{3}{8} = 33\frac{1}{2}$. $4 \times \frac{1}{16} = \frac{1}{4}$ more.

6. **$1\frac{1}{16}$ in**

   $\frac{4}{16} + \frac{10}{16} + \frac{3}{16} = \frac{17}{16}$

7. **Along: $7\frac{1}{2}$ in. Across: $2\frac{1}{4}$ in.**

   $9\frac{3}{4} - 2 \times 1\frac{1}{8}$ and $4\frac{1}{2} - 2 \times 1\frac{1}{8}$

Exercise 4D

1. **$\frac{11}{16}$**

   $\frac{6}{16} + \frac{5}{16}$

2. **$4\frac{5}{8}$**

   $3 + \frac{6}{8} + \frac{7}{8} = 3 + \frac{13}{8}$

3. **$2\frac{11}{16}$**

   $4\frac{17}{16} - 2\frac{6}{16}$

4. **$3\frac{11}{16}$**

   $6\frac{16}{16} - 3\frac{5}{16}$

5. **$\frac{1}{2}$**

   $\frac{6}{12}$

6. **4**

   $\frac{5}{2} \times \frac{8}{5} = \frac{40}{10}$

7. **$14\frac{1}{2}$**

   $4 \times \frac{29}{8} = \frac{116}{8}$

8. **$2\frac{1}{2}$**

   $\frac{5}{8} \times \frac{4}{1} = \frac{20}{8}$

9. **6**

   $\frac{21}{2} \times \frac{4}{7} = \frac{84}{14}$

10. **$2\frac{1}{4}$**

    $\frac{27}{4} \times \frac{1}{3} = \frac{27}{12}$

Exercise 4K

1. **A = $70\frac{9}{16}$ in, B = $71\frac{1}{4}$ in, C = $72\frac{13}{16}$ in**

2. **$13\frac{1}{16}$ in**

   $120 - (36\frac{3}{8} + 42\frac{11}{16} + 27\frac{1}{2}) - 3 \times \frac{1}{8} = 120 - 106\frac{9}{16} - \frac{3}{8}$

3. **4 blocks, offcut $5\frac{1}{2}$ in**

   Each block uses $22\frac{5}{8}$ in. $96 \div 22\frac{5}{8} = 4\frac{44}{181}$

4. **$3\frac{1}{4}$ in**

   $96 - 92\frac{5}{8} - \frac{1}{8}$

5. **$72\frac{11}{16}$ in from either end**

6. **$\frac{13}{16}$ in. $\frac{11}{16}$ in is bigger by $\frac{1}{16}$ in.**

7. **35 mm**

8. **$78\frac{1}{3}$ mm deep, within $117\frac{1}{2}$ mm of the bearing edge**

9. **Loadbearing: $29\frac{2}{3}$ mm. Non-loadbearing: 49 mm.**

   $89 - \frac{2}{3} \times 89 = 89 - 59\frac{1}{3}$. $89 - 40 = 49$.

10. **$31\frac{1}{2}$ mm**

5 Decimals

ANSWERS

Part 1: Textbook

Try it 5.1

1. Three and six hundredths
2. 5 thousandths (0.005)
3. 3
4. 0.9
5. 0.45
6. $\frac{125}{1{,}000}$

Try it 5.2

1. 0.6
2. 0.125, 0.25, 0.5
3. 6.4
4. 9.53
5. 3.0 mm
6. 19 mm

Try it 5.3

1. 5.25
2. 22.225
3. 6.75
4. 57.15
5. 219 mm
6. 0.85 mm

Try it 5.4

1. 0.25
2. 7.125
3. 0.18
4. 475
5. 609.6 mm
6. 15.875 mm

Try it 5.5

1. 2.4
2. 5
3. 50.8
4. 6.35
5. 152.4 mm
6. 33.33 mm

Try it 5.6

1. 0.75
2. 1.875 in
3. $\frac{5}{8}$
4. $3\frac{3}{8}$ in (0.4 × 16 = 6.4, rounds to 6)
5. $\frac{19}{32}$ in (0.59 × 32 = 18.88, rounds to 19)
6. $3\frac{15}{16}$ in (100 ÷ 25.4 × 16 = 62.99, rounds to 63)

Try it 5.7

1. $14.74
2. $18.75
3. $2.16
4. $13.18
5. $12.35
6. $13.68

Try it 5.8

1. 0.875
2. 3.333
3. 0.17
4. 3.143
5. 0.643
6. 0.778

Try it 5.9

1. 10 ft 3 in
2. 6.75 ft
3. 3 ft $4\frac{13}{16}$ in (0.4 × 12 = 4.8 in, 0.8 × 16 = 12.8, rounds to 13)
4. 9.583 ft
5. 2.87 mm
6. $77.92

Worksheet 5A

1. **88.9 mm**
   165.1 − 31.75 − 44.45 = 88.9
2. **A = 13 mm, B = 28.5 mm, C = 36.5 mm**
3. **1,057.2 mm**
   2,438.4 − (457.2 + 609.6 + 304.8) − 3 × 3.2 = 1,057.2
4. **52.5 mm**
   60.3 − 2 × 3.9 = 52.5
5. **457.2 mm**
   152.4 + 76.2 + 152.4 + 76.2
6. **3.05 mm. It is more than 3 mm.**
7. **186.27 mm**
   (1,219.2 − 2 × 50.8) ÷ 6 = 1,117.6 ÷ 6 = 186.27

Worksheet 5B

1. **4.375 in = $4\frac{3}{8}$ in, 2.625 in = $2\frac{5}{8}$ in**
2. **2.875 in = $2\frac{7}{8}$ in**
   6.5 − 1.25 − 2.375 = 2.875
3. **$1\frac{11}{16}$ in (0.71 × 16 = 11.36, rounds to 11) and $1\frac{23}{32}$ in (0.71 × 32 = 22.72, rounds to 23)**
4. **$2\frac{15}{16}$ in**
   75 ÷ 25.4 × 16 = 47.24, rounds to 47
5. **7.94 mm**
   5 ÷ 16 × 25.4 = 7.9375
6. **$79.26**
   3 × 6 × 4.15 + 12 × 0.38 = 74.70 + 4.56
7. **14.25 in = $14\frac{1}{4}$ in. New total 13.5 in = $13\frac{1}{2}$ in.**

Worksheet 5C

1. **11 ft 6 in**
   0.5 × 12 = 6
2. **7.667 ft**
   8 ÷ 12 = 0.6666..., rounds to 0.667
3. **5 ft 3 $\frac{5}{8}$ in**
   0.3 × 12 = 3.6 in. 0.6 × 16 = 9.6, rounds to 10. $\frac{10}{16} = \frac{5}{8}$
4. **2.143 ft**
   15 ÷ 7 = 2.1428..., rounds to 2.143
5. **5.333 ft = 5 ft 4 in**
   16 ÷ 3 = 5.333... 0.333... × 12 = 4 in
6. **$645.70**
   30 × 4.87 = 146.10. 8 × 62.45 = 499.60.
7. **12.2 m, $38.43**
   3.66 + 2.44 + 3.66 + 2.44 = 12.2 m. 12.2 × 3.15
8. **2.64 mm, 3.25 mm, 4.88 mm. 2.24 mm bigger.**

Part 2: Exercise Book

Exercise 5A

1. **4 hundredths (0.04). Seventy-five thousandths.**
2. **0.03 and 0.048**
3. **A = 8.5 mm, B = 26 mm, C = 44.5 mm, D = 57 mm**
4. **0.375, 0.4, 0.41**
5. **2.5 mm, by 0.05 mm**
6. **7.8 and 7.85**
7. **31.5 mm and 19 mm**
   31.27 × 2 = 62.54, rounds to 63, ÷ 2 = 31.5. 18.8 × 2 = 37.6, rounds to 38, ÷ 2 = 19
8. **73.83**
9. **32.54**
10. **(a) 0.24 (b) 177.8 (c) 3,125**
11. **(a) 70 (b) 7.62 (c) 0.195**
12. **0.875 and 3.25**
13. **$\frac{7}{8}$ and $\frac{2}{25}$**
    $\frac{875}{1{,}000} \div 125 = \frac{7}{8}$. $\frac{8}{100} \div 4 = \frac{2}{25}$
14. **$1\frac{1}{2}$ in (0.47 × 16 = 7.52, rounds to 8) and $1\frac{15}{32}$ in (0.47 × 32 = 15.04, rounds to 15)**
15. **$\frac{5}{16}$ in (0.29 × 16 = 4.64, rounds to 5)**
16. **$1\frac{3}{16}$ in**
    30 ÷ 25.4 × 16 = 18.90, rounds to 19. $\frac{19}{16} = 1\frac{3}{16}$
17. **31.75 mm**
    1.25 × 25.4

Exercise 5B

1. **142.75 mm**
   1,500 – (412.5 + 287.25 + 650) – 3 × 2.5 = 1,500 – 1,349.75 – 7.5
2. **$81.03**
   3 × 1.85 = 5.55 m. 5.55 × 14.60
3. **7 shims**
   1.75 ÷ 0.25 = 175 ÷ 25
4. **$14.50**
5. **10.62 kg**
6. **117.86 mm**
   (900 – 2 × 37.5) ÷ 7 = 825 ÷ 7
7. **$3\frac{3}{8}$ in**
   0.38 × 16 = 6.08, rounds to 6. $\frac{6}{16} = \frac{3}{8}$
8. **$4\frac{3}{4}$ in**
   120 ÷ 25.4 × 16 = 75.59, rounds to 76. $\frac{76}{16} = 4\frac{3}{4}$
9. **$802.50**
10. **15.875 mm**
    0.625 × 25.4

Exercise 5C

1. **102.25 mm**
   210 – 45.5 – 62.25
2. **88.9 mm**
   77.9 + 2 × 5.5
3. **101.6 mm**
   (609.6 – 2 × 203.2) ÷ 2
4. **45 mm**
   (750 – 4 × 165) ÷ 2
5. **1,143 mm, then 1,135.5 mm**
   5 × 228.6. 5 × 1.5 = 7.5 mm less.
6. **5.875 in = $5\frac{7}{8}$ in, 3.25 in = $3\frac{1}{4}$ in**
7. **A = 23.5 mm, B = 41 mm, 17.5 mm apart**

Exercise 5D

1. **16.445**

12.070 + 3.500 + 0.875

2. **12.616**

20.000 − 7.384

3. **0.9**

36 × 25 = 900, three decimal places: 0.900

4. **35.4**

1,475 × 24 = 35,400, three decimal places: 35.400

5. **24**

Move both points one place: 96 ÷ 4

6. **3.25**

Start with 19. 6 goes into 19 3 times: 3 × 6 = 18, and 19 − 18 = 1. Write the point in the answer. Bring down the 5 to make 15. 6 goes into 15 twice: 2 × 6 = 12, and 15 − 12 = 3. Write a 0 at the end of the number and bring it down to make 30. 6 goes into 30 5 times: 5 × 6 = 30, and 30 − 30 = 0.

7. **0.625**

Start with 5. 8 goes into 5 zero times. Write 0. Write the point in the answer. Write a 0 at the end of the number and bring it down to make 50. 8 goes into 50 6 times: 6 × 8 = 48, and 50 − 48 = 2. Write a 0 at the end of the number and bring it down to make 20. 8 goes into 20 twice: 2 × 8 = 16, and 20 − 16 = 4. Write a 0 at the end of the number and bring it down to make 40. 8 goes into 40 5 times: 5 × 8 = 40, and 40 − 40 = 0.

8. **2.333**

Start with 7. 3 goes into 7 twice: 2 × 3 = 6, and 7 − 6 = 1. Write the point in the answer. Write a 0 at the end of the number and bring it down to make 10. 3 goes into 10 3 times: 3 × 3 = 9, and 10 − 9 = 1. Write a 0 at the end of the number and bring it down to make 10. 3 goes into 10 3 times: 3 × 3 = 9, and 10 − 9 = 1. Write a 0 at the end of the number and bring it down to make 10. 3 goes into 10 3 times: 3 × 3 = 9, and 10 − 9 = 1. Write a 0 at the end of the number and bring it down to make 10. 3 goes into 10 3 times: 3 × 3 = 9, and 10 − 9 = 1. To three places: 2.333.

9. **0.22 and 0.222**

Start with 2. 9 goes into 2 zero times. Write 0. Write the point in the answer. Write a 0 at the end of the number and bring it down to make 20. 9 goes into 20 twice: 2 × 9 = 18, and 20 − 18 = 2. Write a 0 at the end of the number and bring it down to make 20. 9 goes into 20 twice: 2 × 9 = 18, and 20 − 18 = 2. Write a 0 at the end of the number and bring it down to make 20. 9 goes into 20 twice: 2 × 9 = 18, and 20 − 18 = 2. Write a 0 at the end of the number and bring it down to make 20. 9 goes into 20 twice: 2 × 9 = 18, and 20 − 18 = 2.

10. **0.429**

Start with 3. 7 goes into 3 zero times. Write 0. Write the point in the answer. Write a 0 at the end of the number and bring it down to make 30. 7 goes into 30 4 times: 4 × 7 = 28, and 30 − 28 = 2. Write a 0 at the end of the number and bring it down to make 20. 7 goes into 20 twice: 2 × 7 = 14, and 20 − 14 = 6. Write a 0 at the end of the number and bring it down to make 60. 7 goes into 60 8 times: 8 × 7 = 56, and 60 − 56 = 4. Write a 0 at the end of the number and bring it down to make 40. 7 goes into 40 5 times: 5 × 7 = 35, and 40 − 35 = 5. To three places: 0.429.

Exercise 5K

1. **9 ft 9 in**

0.75 × 12 = 9

2. **12.333 ft**

4 ÷ 12 = 0.3333...

3. **7 ft 10 $\frac{3}{16}$ in**

0.85 × 12 = 10.2 in. 0.2 × 16 = 3.2, rounds to 3.

4. **3.571 ft, 3 ft 6 $\frac{7}{8}$ in**

25 ÷ 7 = 3.5714... 0.5714... × 12 = 6.857 in. 0.857 × 16 = 13.7, rounds to 14 sixteenths.

5. **2.01 mm**

6. **54.9 m, $145.49**

18 × 3.05 = 54.9. 54.9 × 2.65 = 145.485, rounds to 145.49.

7. **$766.47**

682.50 + 83.97

8. **$33.38**

2 × (4.27 + 3.35) = 15.24 m. 15.24 × 2.19 = 33.3756

9. **0.257 m = 257 mm**

6 Percent, Ratio and Proportion

ANSWERS

Part 1: Textbook

Try it 6.1

1. 0.42
2. 0.075
3. 6%
4. 125%
5. $\frac{2}{5}$
6. 62.5%

Try it 6.2

1. 90
2. 100
3. 48
4. 75
5. 20 bolts
6. 609.6 mm

Try it 6.3

1. 25%
2. 3%
3. 15%
4. 120
5. 300
6. 25%

Try it 6.4

1. 69 m
2. 27 sheets (26.4 rounded up)
3. $200.00
4. 1,260 mm
5. 15%
6. 20%

Try it 6.5

1. 2 : 3
2. 3 : 1
3. 1 : 3
4. 600 mL and 300 mL
5. 1,500 mL and 500 mL
6. 5 : 3

Try it 6.6

1. 150 mm per minute
2. 12
3. 24 minutes
4. B: $6.90 per m (A is $7.20 per m)
5. $130.50
6. 3

Try it 6.7

1. Yes. 3 × 20 = 4 × 15 = 60
2. x = 18
3. x = 16
4. 21 m
5. 10 hours
6. 0.6 L

Try it 6.8

1. 480 mm
2. 1,800 mm
3. 60 mm
4. 60 mm
5. 75 : 1,500 = 1 : 20
6. 11 ft

Try it 6.9

1. $819.00
2. $42.00
3. GST $4.00, PST $5.60, total $89.60
4. $135.00, total $151.20
5. $887.60
6. $7.99

Try it 6.10

1. 8 hours
2. 3 days
3. 60 minutes
4. 6 workers
5. Direct
6. Inverse

Try it 6.11

1. **1 in 2**
2. **3 in 12**
3. **No. 3 in 12 is 1 in 4, which is flatter than 1 in 3.**
4. **1,400 mm**
5. **32.2 m²**
6. **3.8 m**

Worksheet 6A

1. **35%, 0.35, $\frac{7}{20}$**
2. **500 mm**
3. **138 m**
   120 × 1.15
4. **25% decrease**
5. **900 mm, then 720 mm. It is not back to 750 mm.**

Worksheet 6B

1. **1,800 mL of base and 600 mL of hardener**
   2,400 mL ÷ 4 parts = 600 mL per part
2. **960 mm by 600 mm**
3. **90 mm**
   $\frac{40}{30} = \frac{120}{x}$, so x = 30 × 120 ÷ 40
4. **96 mm. Every line is twice as long as on the 1 : 20 drawing.**
5. **13 ft**
   3.25 ÷ 0.25
6. **125 L per minute, 18 minutes**

Worksheet 6C

1. **1,250 mm**
   $\frac{5}{12} = \frac{x}{3{,}000}$
2. **1 in 2. Yes, it is steeper than 1 in 3.**
3. **57.2 m²**
   52 × 1.1
4. **4,800 mm by 3,500 mm, 16.8 m²**
   4.8 m × 3.5 m
5. **11 ft**
   2.75 ÷ 0.25
6. **1,200 mm, then 2,400 mm. The rise doubles.**
7. **71 sheets of subfloor and 124 sheets of sheathing**
   64 × 1.1 = 70.4, 112 × 1.1 = 123.2, round up
8. **The 4 in 12 roof, by 0.5 ft (5 ft and 4.5 ft)**
   15 × 4 ÷ 12 and 9 × 6 ÷ 12
9. **10 hours, 5 framers**
   2 × 15 = 30 framer-hours

Part 2: Exercise Book

Exercise 6A

1. **0.64**
2. **0.035**
3. **70%**
4. **240%**
5. $\frac{9}{20}$
6. **87.5%**
7. **78**
8. **273**
9. **7.5%**
10. **120**
11. **391**
12. **1,150**
13. **25%**
14. **3 : 7**
15. **1 : 6**
    250 : 1,500, divide both by 250
16. **750 and 450**
    1,200 ÷ 8 = 150 per part
17. **70 L per minute**
18. **x = 12**
19. **3,400 mm**
20. **42 mm**

Exercise 6B

1. **71.68 m, 12 lengths**
   64 × 1.12 = 71.68, ÷ 6.4 = 11.2, round up
2. **21 sheets (20.7 rounded up)**
3. **$76.12 each, $456.72 for 6**
   86.5 – 12% = 76.12, × 6
4. **0.4 m per s, 125 s**
5. **7.06%**
6. **24.5 m**
   9.8 × 35 ÷ 14
7. **1 L and 0.4 L**
   1.4 ÷ 7 = 0.2 L per part
8. **43 ft**
   5.375 ÷ 0.125
9. **15%**
10. **5 : 2**
    300 : 120, divide both by 60

Exercise 6C

1. **62%, 0.62,** $\frac{31}{50}$
2. **560 mm**
3. **2,640 mm, 55%**
4. **60 mm**
   $\frac{50}{20} = \frac{150}{x}$
5. **1,400 mm by 800 mm**
6. **0.6 L of A and 0.9 L of B**
7. **3.75 ft = 3′ 9″**
8. **80 mm × 50 mm at 1 : 10, 160 mm × 100 mm at 1 : 5. Every line is twice as long.**

Exercise 6D

1. **1.25**
2. **3.4**
3. **0.005**
4. **208%**
5. **1.5%**
6. **150%**
7. **260%**
8. **15%**
9. $1\frac{3}{5}$
10. $2\frac{1}{4}$
11. $\frac{1}{8}$
12. **0.36 and 36%**
13. **345**
14. **25**
15. **41**
16. **420**
17. **84**
    24 + 24 + 24 + 12
18. **96**
    64 + 32
19. **700 L**
    500 + 40% of 500 = 500 + 200
20. **150%**
    $\frac{18}{12} = 1\frac{1}{2} = 150\%$

Exercise 6E

1. **$931.00**
2. **$1,313.20**
   40 × 26.80 = $1,072.00, overtime 6 × 40.20 = $241.20
3. **$952.85**
4. **GST $9.25, PST $12.95, total $207.20**
5. **GST $3.25, PST $4.55, total $72.79**
   GST 64.99 × 0.05 = 3.2495, PST 64.99 × 0.07 = 4.5493
6. **$399.84**
   Sale price $357.00, GST $17.85, PST $24.99
7. **Large can $11.90 per L, small can $16.37 per L. The large can is the better buy.**
8. **8.5¢ and 7.9¢ per screw. The box of 250 costs less per screw.**
9. **$481.50**
10. **$86.81**
11. **10%**
12. **$347.03**
    Before tax $309.85, GST $15.49, PST $21.69

Exercise 6K

1. **1 in 2.4. Yes. 1 in 2.4 is steeper than 1 in 3.**
   12 ÷ 5 = 2.4. A smaller second number means a steeper roof.
2. **1 in 4. Not in normal application. Yes in low slope application.**
   12 ÷ 3 = 4. 1 in 4 is flatter than 1 in 3 but steeper than 1 in 6.
3. **4 in 12**
   $\frac{1}{3} = \frac{x}{12}$
4. **1,200 mm**
   $\frac{4}{12} = \frac{x}{3{,}600}$
5. **7 ft**
6. **47 sheets (46.2 rounded up)**
7. **39.42 m², 18 boxes**
   36.5 × 1.08 = 39.42, ÷ 2.2 = 17.92, round up
8. **6,400 mm = 6.4 m**
9. **96 mm**
10. **13 ft by 11 ft**
11. **0.15 m³ cement, 0.3 m³ sand, 0.45 m³ gravel**
    0.9 ÷ 6 parts = 0.15 m³ per part
12. **6 hours**
    4 × 9 = 36 framer-hours, ÷ 6

7 Measurement and Unit Conversion

ANSWERS

Part 1: Textbook

Try it 7.1

1. 3,600 mm
2. 0.85 m
3. 420 mm
4. 6.096 m
5. 2,500 mL
6. 1.8 t

Try it 7.2

1. 67 in
2. $122\frac{1}{4}$ in
3. 7′ 2″
4. $10′\ 5\frac{1}{2}″$
5. 12 ft
6. $8′\ 4\frac{1}{4}″$

Try it 7.3

1. 304.8 mm
2. 7.9375 mm
3. 6.096 m
4. 30 in
5. $78\frac{3}{4}″$
6. 49.90 kg

Try it 7.4

1. 3,000,000 mm²
2. 0.45 m²
3. 720 in²
4. 6,451.6 mm²
5. 480 L
6. 3,456 in³

Try it 7.5

1. 37.85 L
2. 45.46 L
3. 95.10 US gal
4. 1.201 US gal
5. 7.48 US gal
6. 109.98 imp gal

Try it 7.6

1. 68 °F
2. 14 °F
3. 100 °C
4. 10 °C
5. 204.4 °C
6. 27 °F

Try it 7.7

1. 4.5×10^4
2. 6×10^{-3}
3. 720,000
4. 1.5 mm
5. 10^6 mm²
6. $2.4 \times 10^3 = 2{,}400$

Try it 7.8

1. 6.5 ft
2. 8.25 ft
3. 4.63 ft
4. 9′ 9″
5. $15′\ 4\frac{13}{16}″$
6. $22′\ 2\frac{3}{16}″$

Try it 7.9

1. $7\frac{1}{16}″$
2. $35\frac{7}{16}″$
3. 10 board feet
4. 64.00 board feet
5. 186.00 ft²
6. 2.97 m²

Worksheet 7A

1. **1,219.2 mm by 762 mm**
2. **24″ by 14″**
   610 ÷ 25.4 × 16 = 384.25 sixteenths, rounds to 384.
   355 ÷ 25.4 × 16 = 223.62 sixteenths, rounds to 224.
3. **11′ 9$\frac{1}{4}$″ = 3,587.75 mm**
4. **Length 1,981.2 mm, width 63.5 mm, thickness 9.525 mm**
5. **68.04 kg**
6. **2.4 m, 240 cm, 7′ 10$\frac{1}{2}$″**
   2,400 ÷ 25.4 × 16 = 1,511.81 sixteenths, rounds to 1,512.
   1,512 ÷ 16 = 94.5 in

Worksheet 7B

1. **200 L**
2. **18.70 US gal**
   30 × 12 × 12 = 4,320 in³, ÷ 231
3. **1,080,000 mm² = 1.08 m²**
   2 × (600 × 400 + 600 × 300 + 400 × 300)
4. **302 °F. A drop of 72 °F.**
5. **1 L, then 8 L (8 times as much)**

Worksheet 7C

1. **40 °C = 104 °F, and −22 °F = −30 °C**
2. **71.6 °F**
3. **218.3 °C**
4. **−13 °F**
5. **−15 °C**
6. **32.4 °F**
7. **25 °C**
8. **38 °C is hotter. It is 100.4 °F, 0.4 °F more.**
9. **50 °F to 89.6 °F**
10. **−40. 1.8 × (−40) + 32 = −40**
11. **122 °F, then 140 °F. Up 18 °F.**
12. **−17.8 °C**
13. **15.6 °C**
14. **356 °F**
15. **98.6 °F**
16. **2.2 °C**
17. **45 °C, which is 81 °F**

Worksheet 7D

1. **17.28 m² = 186.00 ft²**
2. **26.67 board feet each, 800 board feet in 30**
   2 × 10 × 16 ÷ 12, where 192 in = 16 ft
3. **7$\frac{1}{16}$″**
   180 ÷ 25.4 × 16 = 113.39 sixteenths, round down to 113
4. **6′ 4$\frac{13}{16}$″**
   1,950 ÷ 25.4 × 16 = 1,228.35 sixteenths, round up to 1,229
5. **34$\frac{1}{16}$″ to 42$\frac{1}{8}$″**
   865 ÷ 25.4 × 16 = 544.88, round up.
   1,070 ÷ 25.4 × 16 = 674.02, round down
6. **14′ 7$\frac{13}{16}$″**
   0.65 × 12 = 7.8 in, 0.8 × 16 = 12.8, about 13 sixteenths
7. **22′ 0$\frac{1}{4}$″ = 22.02 ft = 6,711.95 mm**
   148.5 + 115.75 = 264.25 in
8. **1,472 ft² = 136.75 m²**
9. **4$\frac{15}{16}$″**
   125 ÷ 25.4 × 16 = 78.74 sixteenths, round up to 79
10. **13$\frac{15}{16}$″**
    355 ÷ 25.4 × 16 = 223.62 sixteenths, round down to 223
11. **947 board feet**
    40 × 2 × 6 × 12 ÷ 12 = 480, 20 × 2 × 10 × 14 ÷ 12 = 466.67
12. **9.44 ft**
    9 + 5.25 ÷ 12

Part 2: Exercise Book

Exercise 7A

1. **4,750 mm**
2. **0.625 m**
3. **380 mm**
4. **3.25 L**
5. **2,400 kg**
6. **0.75 kg**
7. **101 in**
8. **$115\frac{3}{4}$ in**
   9 × 12 + 7.75 = 115.75
9. **7′ 10″**
10. **13′ $1\frac{1}{2}$″**
11. **15.875 mm**
12. **57.15 mm**
13. **$37\frac{3}{8}$″**
    950 ÷ 25.4 × 16 = 598.43 sixteenths, rounds to 598
14. **4.572 m**
15. **34.02 kg**
16. **18 ft, 5.4864 m**
17. **2,500,000 mm²**
18. **864 in²**
19. **16,129 mm²**
20. **96 L**
21. **75.71 L**
22. **113.65 L**
23. **95 °F**
24. **−20 °C**

Exercise 7B

1. **7′ 4″ (88″)**
   2,235 ÷ 25.4 × 16 = 1,407.87 sixteenths, rounds to 1,408 = 88 in
2. **14′ $9\frac{1}{4}$″ = 4,502.15 mm**
   54.5 + 87.75 + 35 = 177.25 in
3. **119.75 kg**
4. **27.90 m² = 300.31 ft², buy 1 US gal**
   27.9 ÷ 0.092 903 04, then ÷ 350 = 0.858
5. **243 L = 64.19 US gal**
6. **36.03 US gal**
   30 × 4.546 09 ÷ 3.785 411 784
7. **93.3 °C and 248 °F**
8. **63 °F**
9. **90 in³ = 1,474.84 cm³**
10. **2.9768 m² = 32.04 ft²**

Exercise 7C

1. **914.4 mm by 609.6 mm**
2. **Hole 20.6375 mm, width 101.6 mm, thickness 6.35 mm**
3. **14′ $0\frac{1}{2}$″ = 4,279.9 mm**
   50.5 + 81.25 + 36.75 = 168.5 in
4. **300 mm**
   150 L = 150,000,000 mm³, ÷ (1,000 × 500)
5. **432 in² = 3 ft²**
6. **9 ft² = 1,296 in²**
7. **70 °C = 158 °F, and −4 °F = −20 °C**
8. **0.125 L, then 1 L, 8 times as much**

Exercise 7D

1. **5.75 ft**
2. **11.33 ft**
3. **7.88 ft**
4. **3.19 ft**
   3 + (2 + 5 ÷ 16) ÷ 12
5. **8′ 6″**
6. **6′ 3″**
7. **10′ $9\frac{5}{8}$″**
   0.8 × 12 = 9.6 in, 0.6 × 16 = 9.6, about 10 sixteenths
8. **23′ $5\frac{5}{8}$″**
   0.47 × 12 = 5.64 in, 0.64 × 16 = 10.24, about 10 sixteenths
9. **5.2×10^{7}**
10. **7.5×10^{-4}**
11. **9,100**
12. **$2 \times 10^{3} = 2{,}000$**
13. **59 °F**
14. **22.2 °C**
15. **−0.4 °F**
16. **662 °F**
17. **537.8 °C**
18. **39.6 °F**
19. **15 °C**
20. **181.84 L, a rise of 90 °F**

Exercise 7K

1. **$7\frac{13}{16}$″**
   200 ÷ 25.4 × 16 = 125.98 sixteenths, round down to 125
2. **$4\frac{15}{16}$″**
   125 ÷ 25.4 × 16 = 78.74 sixteenths, round up to 79
3. **$11\frac{1}{16}$″**
   280 ÷ 25.4 × 16 = 176.38 sixteenths, round up to 177
4. **$42\frac{3}{16}$″**
   1,070 ÷ 25.4 × 16 = 674.02 sixteenths, round up to 675
5. **Yes. 5′ 11″ = 71 in = 1,803.4 mm, which is more than 1,800 mm.**
6. **6′ $8\frac{3}{4}$″**
   2,050 ÷ 25.4 × 16 = 1,291.34 sixteenths, round up to 1,292
7. **32 board feet, 576 board feet**
8. **200 board feet**
9. **1.359 m³**
10. **22.68 m² = 244.13 ft²**
11. **252 ft² = 23.41 m²**
12. **13 boxes**
    244.13 ÷ 20 = 12.21, round up

8 Formulas and Transposing

ANSWERS

Part 1: Textbook

Try it 8.1

1. V
2. π (the 2 in r^2 is a power, it tells you to square r)
3. C and D
4. 2 × l × w
5. s × s
6. 1.8 and 32

Try it 8.2

1. 2.88 m^2
2. 2,100 mm
3. 942.48 mm
4. 1,256.64 cm^2
5. 1.5 m^3
6. 95 °F

Try it 8.3

1. 82
2. 200
3. 250
4. 140
5. 4,725
6. 250

Try it 8.4

1. l = A ÷ w
2. 2.4 m
3. 48 in (4 ft)
4. 300 mm
5. 84 mm
6. 0.5 m
7. 600 mm

Try it 8.5

1. 700 mm
2. w = P ÷ 2 − l
3. 100 °C
4. −40 °C
5. 8 hours, from h = (T − 60) ÷ 85
6. 100 mm

Try it 8.6

1. 40 cm
2. 0.6 m
3. 100 mm
4. 50.46 mm
5. 1.91 m, from h = V ÷ (π × r^2)
6. The side becomes 3 times as long (√9 = 3).

Try it 8.8

1. b − 75
2. 0.85 × n + 12
3. 6,000 ÷ k
4. x + 1,980 = 2,750, so x = 770 mm
5. 6 × s = 14,640, so s = 2,440 mm
6. 3 × x + 250 = 1,600, so x = 450 mm
7. 90 + 110 × h = 640, so h = 5 hours
8. L ÷ 2 − 3 = 1,497, so L = 3,000 mm

Try it 8.9

1. 24 V
2. 60 W
3. R = V ÷ I = 12 Ω
4. I = V ÷ R = 2 A
5. V = P ÷ I = 12 V
6. 24 W

Try it 8.10

1. 13 risers of 188.46 mm
2. 3,360 mm
3. 16 risers of 180 mm (2,880 ÷ 180 is exactly 16)
4. 2,790 mm
5. 255 mm. Yes, it is exactly the 255 mm minimum.
6. No. They differ by 7 mm, more than 5 mm.

Worksheet 8A

1. **60 cm**
   w = A ÷ l = 7,200 ÷ 120
2. **45 cm**
   s = √A = √2,025
3. **r = 39.89 mm, D = 79.79 mm**
   r = √(5,000 ÷ π), D = 2 × r
4. **240 mm**
   h = 2 × A ÷ b = 2 × 43,200 ÷ 360
5. **0.75 m**
   h = V ÷ (l × w) = 0.72 ÷ (1.2 × 0.8)
6. **1.77 m**
   h = V ÷ (π × r^2) = 2 ÷ (π × 0.6^2)
7. **90 cm. The side doubled.**
8. **30 cm. The width was halved.**
9. **l × w must be in brackets: h = V ÷ (l × w) = 0.75 m.**
   Check: 1.2 × 0.8 × 0.75 = 0.72.

Worksheet 8B

1. **D = C ÷ π = 636.62 mm**
2. **w = P ÷ 2 – l = 900 mm**
3. **b = 2 × A ÷ h = 450 mm**
4. **$t_C = (t_F - 32) \div 1.8 = 232.22$ °C**
5. **h = (T – 60) ÷ 85 = 5.5 hours**

Part 2: Exercise Book

Exercise 8A

1. **Subject V. Variables V, r and h. Constant π.**
2. **3 × a × b and π × r × r**
3. **9 m²**
4. **4,250 mm**
5. **1,413.72 mm**
6. **3,848.45 mm²**
7. **0.81 m³**
8. **14 °F**
   1.8 × (−10) = −18, −18 + 32 = 14
9. **465**
10. **150**
11. **250**
12. **270**
13. **90**
14. **12.25**
15. **l = A ÷ w**
16. **D = C ÷ π**
17. **w = V ÷ (l × h)**
18. **s = P ÷ 4**
19. **l = P ÷ 2 − w**
20. **$t_C = (t_F - 32) \div 1.8$**
21. **b = 2 × A ÷ h**
22. **h = (T − a) ÷ r**
23. **s = √A**
24. **r = √(A ÷ π)**
25. **h = V ÷ (π × r²)**
26. **n = L ÷ (π × D)**

Exercise 8B

1. **0.9 m**
   l = A ÷ w = 0.54 ÷ 0.6
2. **1.2 m**
   h = V ÷ (l × w) = 1.8 ÷ (1.5 × 1.0)
3. **123.61 mm**
   r = √(12,000 ÷ π) = 61.80, D = 2 × r
4. **1,950 mm**
   l = P ÷ 2 − w = 3,200 − 1,250
5. **121.11 °C**
   (250 − 32) ÷ 1.8
6. **36 in = 3 ft**
   s = √1,296
7. **2.26 m**
   h = 4 ÷ (π × 0.75²)
8. **13 days**
   d = (285.50 − 45) ÷ 18.50
9. **$16\frac{11}{16}$″**
   D = 52.375 ÷ π = 16.671 in, 0.671 × 16 = 10.74, round to 11
10. **200 mm**
    b = 2 × A ÷ h
11. **0.282 m**
    r = √(0.25 ÷ π)
12. **The part is 1.04 m², less than the 1.08 m² shelf, but it is 0.65 m deep and the shelf is only 0.6 m deep, so it does not fit.**
13. **96 in = 8 ft**
    l = A ÷ w = 4,608 ÷ 48
14. **31.83 turns**
    150 ft = 1,800 in, n = 1,800 ÷ (π × 18)
15. **Learner: 4,000 − 1,400 = 2,600 mm. Check: 2 × (2,600 + 700) = 6,600, not 4,000. Correct: l = P ÷ 2 − w = 1,300 mm.**

Exercise 8C

1. **60 cm**
2. **56 mm**
3. **r = 50.46 mm, D = 100.93 mm**
4. **180 mm**
5. **1.2 m**
6. **0.44 m**
   r = √(V ÷ (π × h)) = √(0.9 ÷ (π × 1.5))
7. **450**
8. **477.46 mm**
9. **168 mm, three times the side**
10. **201.85 mm. The diameter doubled.**

Exercise 8K

1. **13 risers of 197.31 mm**
   2,565 ÷ 200 = 12.825, round up to 13. 2,565 ÷ 13
2. **Yes, 265 mm is between 255 mm and 355 mm. Total run 12 × 265 = 3,180 mm**
3. **17 risers of 174.12 mm**
   2,960 ÷ 180 = 16.44, round up to 17
4. **2,960 ÷ 16 = 185 mm, more than the 180 mm maximum for a public stair**
5. **2,848 mm**
   total rise = number of risers × unit rise
6. **255 mm. Yes, it is exactly the 255 mm minimum**
   unit run = total run ÷ number of treads
7. **Yes. Adjacent differences are 3, 4, 5, 2 and 5 mm, none more than 5 mm. Tallest minus shortest is 186 – 179 = 7 mm, within 10 mm**
8. **No. 276 – 270 = 6 mm and 276 – 268 = 8 mm are more than 5 mm**
9. **No, 3,900 mm is more than 3,700 mm. Each flight rises 1,950 mm: 10 risers of 195 mm**
10. **190 × n = 2,660, so n = 14**
11. **unit run = total run ÷ number of treads = 3,640 ÷ 14 = 260 mm**
12. **Unit rise 2,280 ÷ 12 = 190 mm. Total run 11 × 250 = 2,750 mm**
13. **12 treads: 3,400 ÷ 12 = 283.33, so 283 mm. It is between 255 mm and 355 mm, so it meets the table**
14. **14 risers of 186.07 mm. 2,605 ÷ 200 = 13.025 rounds up to 14, one more riser, so each riser is lower**

9 Perimeter and Circumference

ANSWERS

Part 1: Textbook

Try it 9.1

1. 3,400 mm
2. 3.6 m
3. 750 mm
4. 640 mm
5. 144 in = 12 ft
6. 500 mm
7. $121\frac{1}{2}$ in

Try it 9.2

1. 90 mm
2. 84.15 mm
3. 12 in
4. The circumference, a little more than 3 times longer (π times).

Try it 9.3

1. 628.32 mm
2. 376.99 mm
3. $31\frac{7}{16}$"
4. 795.77 mm, from D = C ÷ π
5. 400 mm
6. 15,707.96 mm
7. 2,000 mm

Try it 9.4

1. 785.40 mm
2. 1,286.64 mm
3. $14\frac{1}{8}$"
4. 300 mm

Try it 9.5

1. 785.40 mm
2. 314.16 mm
3. 471.24 mm
4. 235.62 mm

Try it 9.6

1. 162.83 mm
2. 1,914.16 mm (four quarter circles make one circle, D = 100 mm)
3. 1,671.24 mm
4. 1,071.24 mm

Try it 9.7

1. 18,300 mm
2. 29,280 mm
3. 33,600 mm
4. 12,390 mm
5. 480 in = 40 ft
6. 9 boards (39,200 ÷ 4,880 = 8.03)

Worksheet 9A

1. 5,500 mm
2. 42 cm
3. 750 mm
   w = P ÷ 2 − l = 4,200 ÷ 2 − 1,350
4. 2 × (124 + 80) = 408 in = 34 ft
5. 500 mm
   s = P ÷ 4
6. 11,000 mm. The perimeter doubled.

Worksheet 9B

1. C = π × 350 = 1,099.56 mm
2. D = 3,000 ÷ π = 954.93 mm
3. π × 400 = 1,256.64 mm
4. 75 ÷ 360 × π × 800 = 523.60 mm
5. 4.55 turns
6. $27\frac{1}{8}$"
7. Curved edge 785.40 mm, perimeter 1,285.40 mm

Worksheet 9C

1. 2 × 80 + π × 40 = 285.66 mm
2. 500 + 500 + 240 + π × 240 ÷ 2 = 1,616.99 mm
3. 2 × 720 + 2 × 420 + π × 80 = 2,531.33 mm
4. π × 120 + π × 60 = 565.49 mm
5. 2 × 1,000 + π × 250 = 2,785.40 mm
6. 2 × 250 + π × 500 ÷ 4 = 500 + 392.70 = 892.70 mm

Part 2: Exercise Book

Exercise 9A

1. **7,100 mm**
2. **300 cm**
3. **3.2 m**
4. **450 mm**
5. **360 in = 30 ft**
6. **$50\frac{3}{4}$ in**
   2 × (15 6/8 + 9 5/8) = 2 × 25 3/8
7. **2 × (111 + 70) = 362 in = 30′ 2″**
8. **700 mm**
9. **136.5 mm**
10. **165 mm**
11. **1,193.81 mm**
12. **596.90 mm**
13. **477.46 mm**
14. **44″**
15. **314.16 mm**
16. **1,570.80 mm**
17. **1,005.31 mm**
18. **It is cut in half too.**

Exercise 9B

1. **2 × (30 + 21) = 102 in = 8′ 6″**
2. **5,962 mm**
   π × 1,850 + 150 = 5,961.95
3. **1,140 mm**
   π × 355 + 25 = 1,140.27
4. **$33\frac{3}{4}$″**
   π × 10.75 = 33.772 in, 0.772 × 16 = 12.35
5. **1,138.83 m**
   1,450 × π × 250 ÷ 1,000
6. **15.92 turns**
   100 ft = 1,200 in, 1,200 ÷ (π × 24)
7. **528.73 mm**
8. **650 mm**
   2,042 ÷ π = 649.99
9. **16,000 mm, so 3 lengths (2 lengths hold only 12,000 mm)**
10. **2,356.19 mm**
    90 ÷ 360 × π × 3,000
11. **Before 3,142.48 mm, after 3,442.48 mm, 300 mm longer (each straight run grows by 150 mm).**
12. **3,872.74 mm**
    π × 1,220 + 40
13. **2,050 mm**
    2 × (600 + 400) + 50

Exercise 9C

1. **1,900 mm**
2. **51 cm**
3. **π × 500 = 1,570.80 mm**
4. **120 ÷ 360 × π × 700 = 733.04 mm**
5. **2 × 600 + π × 250 = 1,985.40 mm**
6. **2 × 780 + 2 × 280 + π × 120 = 2,496.99 mm**
   Straight edges 900 − 2 × 60 = 780 and 400 − 2 × 60 = 280. Four quarter circles make one circle, D = 120
7. **885.84 mm**
   c = (L − π × D) ÷ 2
8. **π × 140 + π × 80 = 691.15 mm (was 659.73 mm)**

Exercise 9K

1. **21,960 mm**
2. **6,100 mm**
3. **146,400 mm, 30 lengths**
   P = 2 × (14,640 + 9,760) = 48,800, × 3, ÷ 4,880
4. **37,820 mm, 11 lengths (10.33 rounded up)**
5. **44,800 mm**
   2 × (12,000 + 1,200 + 8,000 + 1,200)
6. **132 ft**
   18 in = 1.5 ft. 2 × (36 + 3 + 24 + 3)
7. **13,480 mm**
8. **546 in = 45′ 6″**
9. **4,860 mm**
10. **7,320 mm more (3 plates × 2 walls × 1,220 mm)**
11. **450 mm**
    33,600 ÷ 2 = 16,800. 16,800 – (9,000 + 6,000) = 1,800, shared by 4 overhangs
12. **16,170 mm**

10 Area

ANSWERS

Part 1: Textbook

Try it 10.1

1. 28 cm²
2. Thirty-five square metres
3. 12 ft²
4. 30 ft²

Try it 10.2

1. 300,000 mm²
2. 62,500 mm²
3. 15 ft² = 2,160 in²
4. 7 tins (40 ÷ 6 = 6.67 L)

Try it 10.3

1. 30,000 mm²
2. 94.5 in²
3. 0.54 m²
4. 30,000 mm²

Try it 10.4

1. 150,000 mm²
2. 360 in²
3. 80,000 mm²
4. 162 in²

Try it 10.5

1. 20,106.19 mm²
2. 62,831.85 mm²
3. 876.50 mm²
4. 863.94 mm² (R = 30, r = 25)

Try it 10.6

1. 130,000 mm²
2. 99,893.81 mm²
3. 58,743.36 mm²
4. 250,000 mm²

Try it 10.7

1. 760,000 mm²
2. 470,000 mm²
3. 16.00 ft² (2,304 in²)
4. 314,159.27 mm²

Try it 10.8

1. 4 sheets (3.34, round up)
2. 13 sheets (12.19)
3. 11 boxes, $423.50 (10.34 boxes before rounding)
4. 2 cans (40 ÷ 35 = 1.14)
5. 5 sheets (140 ÷ 32 = 4.375)

Worksheet 10A

1. 750,000 mm² = 0.75 m²
2. 720 in² = 5 ft²
3. 280,000 mm²
4. 300 mm
   54,000 × 2 ÷ 360 = 300
5. 159,043 mm²
6. 14,137.17 mm²
   π × (90² – 60²) = π × 4,500

Worksheet 10B

1. B is 450 mm high. Area =
   90,000 + 90,000 = 180,000 mm²
2. Hidden width 600 mm. Area =
   75,000 + 75,000 + 90,000 = 240,000 mm²
3. 312 in² = 2.17 ft²
   240 + 72 = 312 in², 312 ÷ 144
4. 131,415.93 mm²
   A = 500 × 200, B + C = one full circle, π × 100²

Worksheet 10C

1. 1,840,000 mm² = 1.84 m²
2. 31,415.93 mm² and 125,663.71 mm². The new area is 4 times bigger (2 × 2).
3. Stretch-out π × 400 = 1,256.64 mm. SA = 1,005,309.65 mm²
   Ends π × 200² × 2, side π × 400 × 600
4. 52 in across. 2,496 in² = 17.33 ft²

Worksheet 10D

1. **11.52 ÷ 2.88 = 4 exactly, so 4 sheets**
   5.4 × 2.4 − 1.2 × 1.2 = 11.52 m², 11.52 ÷ 2.88
2. **6 m**
   5 × 2.88 = 14.4 m², 14.4 ÷ 2.4
3. **12.96 m², 13.20 boxes, so 14 boxes, $588.00**
   2,400 × 3,600 + 2,400 × 1,800 mm² = 12.96 m², × 1.10 ÷ 1.08
4. **90 m², 29.03 bundles, so 30 bundles, $1,080.00**
   2 × 10 × 4.5 = 90 m², 90 ÷ 3.1
5. **5.25 sheets, so 6 sheets, $164.40**
   24 × 8 − 6 × 4 = 168 ft², 168 ÷ 32
6. **1.91 L, so 2 tins**
   4 × 2.6 − 0.9 × 2 = 8.6 m², × 2 coats = 17.2 m², ÷ 9

Part 2: Exercise Book

Exercise 10A

1. **45 cm²**
2. **1.5 m²**
3. **80,000 mm²**
4. **864 in², 3 ft²**
5. **294,000 mm²**
6. **2.56 m²**
7. **70,200 mm²**
8. **52.5 in²**
9. **0.3 m²**
10. **59,400 mm²**
11. **6,361.73 mm²**
12. **706.86 in²**
    r = 15 in
13. **2.26 m²**
14. **196,350 mm²**
15. **6,283.19 mm²**
    R = 60, r = 40, π × (3,600 − 1,600)
16. **The radii must be squared before subtracting: π × (60² − 40²), not π × (60 − 40)².**
17. **9 in**
    54 × 2 ÷ 12
18. **25,000 cm²**

Exercise 10B

1. **25.92 m², 9 sheets**
   7.2 × 3.6 = 25.92 m², one sheet = 2.88 m², 25.92 ÷ 2.88 = 9
2. **4.57 L, so 5 tins**
   Area 2 × 4 × 4 = 32 m², 32 ÷ 7
3. **52,778.76 mm²**
4. **2,304 in² = 16 ft²**
   16 × 12 ÷ 2 = 96 in² each
5. **200,979.47 mm²**
6. **3.24 m²**
   One side 810,000 mm²
7. **40 ft² (5,760 in²)**
8. **9.80 m²**
   Ends π × 0.6² × 2, side π × 1.2 × 2
9. **504,657.08 mm²**
10. **1.58 m²**
    Stretch-out π × 168, times 3,000
11. **900,000 mm² = 0.9 m²**
12. **133 ft²**
    9 ft 6 in = 9.5 ft, 14 × 9.5

Exercise 10C

1. **700 mm**
   420,000 ÷ 600
2. **500 mm**
   300,000 ÷ ((400 + 800) ÷ 2) = 300,000 ÷ 600
3. **100 mm**
   r = √(31,416 ÷ π)
4. **Foot 700 mm long. Area = 225,000 + 175,000 = 400,000 mm²**
5. **180,000 + 30,000 = 210,000 mm²**
6. **40,796.46 mm²**
7. **1,370,000 mm² = 1.37 m²**
   2 × (360,000 + 225,000 + 100,000)
8. **Stretch-out 942.48 mm. SA = 612,610.57 mm²**
9. **90,000 mm² and 810,000 mm², 9 times bigger (3 × 3)**

Exercise 10D

1. **4.59 sheets, so 5 sheets, $96.25**
   Net area 13.23 m²
2. **4 sheets exactly**
   17.28 m² ÷ 4.32 m²
3. **27.6, so 28 tiles**
   One tile 0.3 × 0.6 = 0.18 m²
4. **198 tiles**
   One tile 0.1 × 0.1 = 0.01 m²
5. **2.30 cans, so 3 cans, $149.97**
   Walls 45 m², net 40.23 m², two coats 80.46 m²
6. **1.54 gal, so 2 gal**
   540 ft² ÷ 350
7. **9.63 boxes, so 10 boxes, $449.00**
8. **10.58, so 11 boxes**
   Net area 22.32 m²
9. **14.7 m², so buy 15 m², $487.50**
10. **Yes. 6 sheets cover 17.28 m², the wall needs 14.85 m², 2.43 m² to spare**
11. **2.72, so 3 sheets**

Exercise 10K

1. **12.7 mm and 15.9 mm (600 mm on walls). 9.5 mm is listed for walls only perpendicular to the framing, at up to 400 mm o.c.**
2. **Parallel: no, the limit is 400 mm. Perpendicular: yes, the limit is 600 mm.**
3. **15.5 mm. 23.22, so 24 sheets**
   69.12 m² ÷ 2.9768 m²
4. **18.5 mm, 3 mm thicker than 15.5 mm**
5. **9.5 mm with edges supported, 12.5 mm with edges unsupported**
6. **31.04, so 32 sheets**
   2 × 11 × 4 = 88 m², × 1.05 = 92.4 m²
7. **4 in 12 = 1 in 3 (12 ÷ 4 = 3). Normal application, at its minimum slope**
8. **3 in 12 = 1 in 4. It is steeper than 1 in 6 but not as steep as 1 in 3, so low slope application only**
9. **2 in 12 = 1 in 6. Low slope application, at its minimum slope. Normal application is not allowed**
10. **Walls 11.91, so 12. Ceiling 5.25, so 6. 18 sheets, $287.64**
    Walls 416 − 35 = 381 ft², ceiling 168 ft²
11. **6.0 mm at 400 mm o.c., 7.5 mm at 600 mm o.c.**
12. **29.06, so 30 sheets**
    96 − 9.5 = 86.5 m²
13. **13.42, so 14 boxes**
    19.2 + 6.4 − 1.2 = 24.4 m²

11 Volume and Capacity

ANSWERS

Part 1: Textbook

Try it 11.1

1. 30 cm^3
2. 1,000 mm^3
3. 3,456 in^3
4. 8 m^3

Try it 11.2

1. 0.12 m^3
2. 5,184 in^3 = 3 ft^3
3. 0.3 m^3
4. 1,000 cm^3

Try it 11.3

1. 2,500 L
2. 200 L
3. 189.27 L
4. 52.83 US gal
5. 4 US gal (924 ÷ 231)
6. 28.32 L

Try it 11.4

1. 254.47 L
2. 78.54 L
3. 70.50 US gal
4. 17.67 L

Try it 11.5

1. 37.70 L
2. 48 L (600^2 × 400 ÷ 3 = 48,000,000 mm^3)
3. 480 in^3
4. 400 mm, the height at right angles to the base

Try it 11.6

1. 4.19 L
2. 56.55 L
3. 0.524 m^3
4. 904.78 in^3

Try it 11.7

1. 1,727.88 cm^3
2. 75.40 L
3. 252 L
4. 80 L (4 times)

Try it 11.8

1. 1.44 m^3 = 1.88 yd^3
2. 11.11 yd^3
3. 0.44 m^3
4. 200 mm wide, 0.24 m^3
5. 23.12 kg

Try it 11.9

1. 266.67 BF
2. 6.67 BF
3. 10.67 BF
4. 213.33 BF, $256.00

Try it 11.10

1. 528.34 US gal, 439.94 imperial gal
2. 149.61 US gal = 566.34 L = 124.58 imperial gal
3. 954.26 L = 252.09 US gal = 209.91 imperial gal
4. 120.09 US gal

Worksheet 11A

1. 0.2 m^3 = 200 L
2. 27,000,000 mm^3 = 27 L
3. 120 L
   600 × 400 ÷ 2 × 1,000
4. 600 mm
   480 L = 480,000,000 mm^3, ÷ (1,000 × 800)
5. 157.08 L
6. 42.41 L

Worksheet 11B

1. 376.99 L
   A = π × 400^2 × 600, B = π × 400^2 × 450 ÷ 3
2. 261.80 L
   Cylinder π × 250^2 × 1,000 plus one sphere 4 ÷ 3 × π × 250^3
3. 33.51 L
4. 9,600 in^3 ÷ 231 = 41.56 US gal

Worksheet 11C

1. **15.71 L and 62.83 L, 4 times bigger**
2. **8 L and 64 L, 8 times bigger**
3. **39.27 L and 157.08 L, 4 times bigger**
   $\pi \times 0.025^2 \times 20 \times 1{,}000$ and $\pi \times 0.05^2 \times 20 \times 1{,}000$
4. **28.27 L**
   The circles are 300 mm across, so r = 150 mm. The height is 400 mm

Part 2: Exercise Book

Exercise 11A

1. **72 cm^3**
2. **3,000 mm^3**
3. **2.5 m^3**
4. **450 L**
5. **2 ft^3**
   3,456 ÷ 1,728
6. **2 L**
7. **0.07 m^3**
8. **0.216 m^3**
9. **45 L**
10. **5 US gal**
11. **283.91 L**
12. **132.09 US gal**
13. **42.41 L**
14. **18.85 L**
15. **162 L**
16. **65.45 L**
17. **0.134 m^3**
18. **5.03 L**
19. **0.6 m**
20. **7.48 US gal**
21. **The 40 L drum. 10 US gal = 37.85 L**
22. **0.063 m^3**

Exercise 11B

1. **180 L, 9 pails**
2. **52.88 US gal**
   r = 9 in, $\pi \times 81 \times 48$ in^3, ÷ 231
3. **159.28 L**
4. **363.25 L**
5. **900 L, 20 minutes**
6. **1,432.57 L**
   Cylinder 1.1310 m^3 + cone 0.3016 m^3
7. **57.66 US gal**
8. **2,094.40 L**
9. **1,099.56 L**
10. **5,890.49 cm^3**
11. **12,214.51 cm^3**
12. **74.81 US gal, about 30 hours**
13. **432 L**

Exercise 11C

1. **800 mm**
   240,000,000 mm^3 ÷ (600 × 500)
2. **1,000 mm**
   $h = V \div (\pi \times r^2)$
3. **64 L**
   $400^3 = 64{,}000{,}000$ mm^3
4. **117.81 L**
   The circles are 500 mm across, so r = 250 mm
5. **179.07 L**
6. **134.04 L**
7. **End area 1,056 mm^2, volume 422.4 cm^3**
   80 × 8 + 52 × 8 = 640 + 416
8. **4.19 L and 33.51 L, 8 times bigger (2 × 2 × 2)**
9. **28.27 L and 56.55 L, 2 times bigger**

Exercise 11D

1. **1.73 m^3 = 2.26 yd^3**
2. **4.44 yd^3**
   4 in = 4 ÷ 12 ft, 120 ft^3 ÷ 27
3. **2.16 m^3**
4. **0.31 m^3**
5. **4,838.4 kg to 5,184 kg**
6. **84.78 kg**
7. **29.59 kg**
8. **298.67 BF, $373.33**
9. **6.67 BF sold, 4.38 BF of wood**
10. **2,400 L = 634.01 US gal = 527.93 imperial gal**
11. **3,534.29 L = 933.66 US gal = 777.44 imperial gal**
12. **2,273.05 L = 600.47 US gal**

Exercise 11K

1. **250 mm, 350 mm and 450 mm**
2. **200 mm, 350 mm and 500 mm. They are the same, 350 mm, for 2 floors**
3. **Projection (350 – 200) ÷ 2 = 75 mm. Thickness is the greater of 100 mm and 75 mm: 100 mm**
4. **Projection (600 – 200) ÷ 2 = 200 mm, so at least 200 mm thick**
5. **1.54 m^3 = 2.01 yd^3**
6. **0.8660 m, so 867 mm**
   side = $\sqrt{0.75}$
7. **0.60 m^3**
8. **75 mm. 6.80 m^3 = 8.90 yd^3**
9. **100 mm. 9.072 m^3**
10. **No. Clause (2)(a) says granular material need not be installed under slabs in garages, carports or accessory buildings**
11. **1.8 + 0.15 = 1.95 m. 14.04 m^3**
12. **41.85 m^3**
    Trench 57.6 m^3, footing 2.43 m^3, wall in the trench 13.32 m^3

12 Angles and Right Triangles

ANSWERS

Part 1: Textbook

Try it 12.1

1. 180°
2. 45°
3. 45°
4. 72°
5. 30°
6. 3/4 of a turn (270 ÷ 360)

Try it 12.2

1. acute, right, obtuse, acute, straight
2. 108°
3. 45°
4. 105°
5. 90°
6. obtuse

Try it 12.3

1. 60°
2. 63°
3. 45°
4. No. Two right angles already add to 180°, leaving 0° for the third angle.
5. 65° each
6. 60°

Try it 12.4

1. 1,000 mm
2. 1,749.29 mm
3. 1,200 mm
4. 1.73 m
5. 15″
6. $28\frac{5}{16}$″ (28.302″)

Try it 12.5

1. 500 mm
2. 15 ft
3. Closed (less than 90°). It should read 1,500 mm.
4. 2 m
5. 25″
6. Yes. $25 + 144 = 169 = 13^2$

Try it 12.6

1. 2,000 mm
2. 1,950 mm
3. 3,354 mm
4. 424.26 mm
5. No. The diagonals are not equal.
6. $87\frac{3}{4}$″

Try it 12.7

1. 62°
2. 152°
3. 76°, 104° and 76°
4. Three more of 41° and four of 139°
5. 73° (alternate angles are equal)
6. 65°

Try it 12.8

1. 25.5°
2. 12.26°
3. 63°45′0″
4. 18°7′30″
5. 37°24′15″
6. 107°18′24″

Try it 12.9

1. Total rise 2,220 mm, total run 2,970 mm, stringer line 3,708.00 mm
2. No on both. The maximum rise is 200 mm for private stairs and 180 mm for public stairs (Table 9.8.4.1).
3. No. The minimum run for a public stair is 280 mm.
4. 260 mm to 285 mm
5. Yes for the dwelling unit (1,950 mm minimum). No for the public stair (2,050 mm minimum).
6. $44\frac{5}{8}$″ (44.598″)

Try it 12.10

1. Rise 1,000 mm, rafter line 3,162.28 mm
2. Run 4,500 mm, rise 3,000 mm, rafter line 5,408.33 mm
3. $169\frac{11}{16}$″, which is 14′ $1\frac{11}{16}$″
4. 13.892″
5. 6,000 mm
6. 23′ $3\frac{7}{8}$″ (279.886″)

Worksheet 12A

1. 58°
2. 63°
3. 42°
4. 115°
5. 45°
6. 72° each
7. Two triangles hold 2 × 180° = 360°. ? = 360 − 85 − 100 − 75 = 100°
8. 125°, an obtuse angle
9. 36° with 10 holes, 30° with 12 holes. More holes give a smaller angle.

Worksheet 12B

1. 750 mm
2. 0.5 m
3. $26\frac{5}{8}$″ (26.627″)
4. 2,012.46 mm
5. 1,750 mm
6. It should read 1,500 mm. At 1,507 mm the corner is open past 90°.
7. 500 mm. With legs of 600 mm and 800 mm it is 1,000 mm, so it doubles too.
8. Bottom diagonal $\sqrt{(600^2 + 400^2)} = 721.11$ mm. Rod $\sqrt{(520{,}000 + 300^2)} = 781.02$ mm

Part 2: Exercise Book

Exercise 12A

1. **120°**
2. **40°**
3. **acute, right, obtuse, obtuse**
4. **116°**
5. **75°**
6. **57°**
7. **52°**
8. **58° each**
9. **110°**
10. **2,500 mm**
11. **594.05 mm**
    $\sqrt{(350^2 + 480^2)} = \sqrt{352{,}900}$
12. **1,500 mm**
    $\sqrt{(1{,}700^2 - 800^2)} = \sqrt{2{,}250{,}000}$
13. **2.19 m**
    $\sqrt{(2.6^2 - 1.4^2)} = \sqrt{4.8}$
14. **$23\frac{5}{8}$″ (23.601″)**
    $\sqrt{(196 + 361)} = \sqrt{557}$
15. **Yes. $64 + 225 = 289 = 17^2$**
16. **No. $36 + 81 = 117$, but $11^2 = 121$**
17. **1,253 mm (1,253.00 mm)**
18. **636.40 mm**
    $450 \times \sqrt{2}$

Exercise 12B

1. **30° between neighbours. Hole 1 to hole 4 is 3 spaces: 90°**
2. **38°, acute**
3. **750 mm each. Two braces and two cuts: $2 \times 750 + 2 \times 3 = 1{,}506$ mm**
   $\sqrt{(450^2 + 600^2)} = \sqrt{562{,}500} = 750$
4. **It should read 2 m. At 1.998 m it is closed, less than 90°.**
5. **3,499 mm each (3,498.57 mm). Push the two corners at the ends of the 3,505 mm diagonal.**
   $\sqrt{(3{,}000^2 + 1{,}800^2)} = \sqrt{12{,}240{,}000}$
6. **2,387.99 mm**
   $\sqrt{(2{,}250^2 + 800^2)} = \sqrt{5{,}702{,}500}$
7. **$91\frac{9}{16}$″, which is 7′ $7\frac{9}{16}$″**
   48″ and 78″: $\sqrt{(2{,}304 + 6{,}084)} = \sqrt{8{,}388} = 91.586$″
8. **$18 + 24 + 30 = 72$″**
   Hypotenuse $\sqrt{(18^2 + 24^2)} = 30$″
9. **120° evenly spaced. The two arms that did not move keep 120° between them, and the third angle is $360 - 100 - 120 = 140°$. The other two angles are 120° and 140°.**

Exercise 12C

1. **390 mm**
2. **1.5 m**
3. **62°**
4. **120°**
5. **57°**
6. **1,700 mm**
7. **721.11 mm. No, it did not double.**
8. **Legs 900 mm and $600 - 300 = 300$ mm. Sloped edge $\sqrt{(900^2 + 300^2)} = 948.68$ mm**
9. **Face diagonal 707.11 mm. Rod $\sqrt{(707.11^2 + 500^2)} = 866.03$ mm**

Exercise 12D

1. **56°**
2. **63°**
3. **133°, 47° and 133° (going round from the 47° angle)**
4. **d = 116°, e = 64°, f = 116°**
5. **58°**
6. **71°**
7. **52.3°**
8. **19.74°**
9. **28°39′0″**
10. **71°15′9″**
    0.2525 × 60 = 15.15′, 0.15 × 60 = 9″
11. **75°18′23″**
    74°77′83″ = 74°78′23″ = 75°18′23″
12. **83°17′45″**
    179°59′60″ − 96°42′15″

Exercise 12E

1. **Total rise 2,820 mm, total run 3,780 mm, stringer line 4,716.02 mm**
   $\sqrt{(2{,}820^2 + 3{,}780^2)}$
2. **13 risers of 196.92 mm**
   2,560 ÷ 200 = 12.8, round up to 13. 2,560 ÷ 13
3. **16 risers of 180 mm, 15 treads, total run 4,200 mm, stringer line 5,092.58 mm**
   2,880 ÷ 180 = 16. 15 × 280 = 4,200. $\sqrt{(2{,}880^2 + 4{,}200^2)}$
4. **Total rise 59″, total run 71.75″, stringer line $92\frac{7}{8}$″ (7′ $8\frac{7}{8}$″)**
   $\sqrt{(59^2 + 71.75^2)} = 92.893$″
5. **Yes. The largest step between neighbours is 4 mm, and 190 − 182 = 8 mm.**
6. **No. 268 mm to 262 mm is 6 mm and 262 mm to 271 mm is 9 mm between adjacent treads (5 mm allowed). The 10 mm check passes: 271 − 262 = 9 mm.**
7. **No. The depth may be at most 255 + 25 = 280 mm.**
8. **2,050.00 mm**

Exercise 12K

1. **Run 4,200 mm, rise 2,100 mm, rafter line 4,695.74 mm**
   4,200 × 6 ÷ 12 = 2,100. $\sqrt{(4{,}200^2 + 2{,}100^2)}$
2. **Rise 52″, rafter line $164\frac{7}{16}$″ (13′ $8\frac{7}{16}$″)**
   156 × 4 ÷ 12 = 52. $\sqrt{(156^2 + 52^2)} = 164.438$″
3. **15″ per foot. 10 × 15 = 150″ = 12′ 6″**
4. **5,161 mm (5,161.40 mm)**
5. **2,700 mm and 3,600 mm from the corner, 4,500 mm apart**
6. **8,357 mm. Push the two corners at the ends of the 8,362 mm diagonal toward each other.**
7. **13 risers of 188.46 mm. Yes.**
   2,450 ÷ 200 = 12.25, round up to 13
8. **No, the public maximum is 180 mm. 2,700 ÷ 180 = 15 risers of 180 mm.**
9. **Yes in a house (1,980 ≥ 1,950). No for a public stair (1,980 < 2,050).**
10. **No. 193 mm to 199 mm is 6 mm between adjacent risers (5 mm allowed). The tallest and shortest differ by 7 mm, which is inside 10 mm.**
11. **Run 3,000 mm, rise 1,000 mm, rafter line 3,162.28 mm**

13 Trigonometry

ANSWERS

Part 1: Textbook

Try it 13.1

1. q
2. p
3. r
4. p
5. q
6. 59°

Try it 13.2

1. 0.6428
2. 0.7660
3. 0.8391
4. 1
5. 0.8660 and 0.8660. The same, because 60° and 30° are the two acute angles of one triangle.
6. sin A = 5 ÷ 13 = 0.3846, tan A = 5 ÷ 12 = 0.4167

Try it 13.3

1. 338.09 mm
2. 1,149.07 mm
3. 649.84 mm
4. 1,046.07 mm
5. 703.16 mm
6. $19\frac{5}{16}$″ (19.284″)

Try it 13.4

1. 32.01°
2. 44.43°
3. 38.21°
4. 36.87° and 53.13°
5. 30°
6. 45°

Try it 13.5

1. 4.29°, 7.50%
2. 2.86°
3. 150 mm
4. 209.27 mm
5. 26.57°
6. 11,449 mm

Try it 13.6

1. 848.53 mm
2. Travel 461.88 mm, run 230.94 mm
3. 653.28 mm
4. $14\frac{1}{8}$″
5. 450 mm
6. 45°

Try it 13.7

1. Corner 108°, cut 54°
2. 12 (each corner turns 30°, and 360 ÷ 30 = 12)
3. 11.55 mm
4. 29.36°
5. 70°
6. 92° (2 × 46°)

Try it 13.8

1. 18.43°
2. 1 in 1.71, 30.26°
3. 1 in 96, 1.04%
4. 2.5 in 12 is 1 in 4.8. Not in normal application (needs 1 in 3, 4 in 12). Yes in low slope application (needs 1 in 6, 2 in 12).
5. Run 2,700 mm, angle 9.46°
6. Seat cut 33.69°, plumb cut 56.31°

Worksheet 13A

1. 321.39 mm
2. 638.05 mm
3. 1.69 m
4. 1,307.59 mm
5. $12\frac{5}{16}$″ (12.313″)
6. $31\frac{3}{8}$″ (31.382″)

Worksheet 13B

1. **33.69°**
2. **36.87°**
3. **30.07°**
4. **16.26° and 73.74°**
5. **Angle 4.76°, grade 8.33%**
6. **Angle 22.62°, sloped edge 13″**

Worksheet 13C

1. **Travel $25\frac{7}{16}$″ (25.456″), run 18″**
2. **Travel 783.94 mm, run 724.26 mm**
3. **29.99°, so 30° fittings**
4. **True offset √(300² + 400²) = 500 mm, travel 707.11 mm**
5. **12.28 mm**
6. **Corner 135°, cut 67.5°**
7. **At 6°: 313.59 mm. At 12°: 623.74 mm. No, it is a little less than double (2 × 313.59 = 627.18).**

Part 2: Exercise Book

Exercise 13A

1. **Opposite M: e. Adjacent to M: d. Hypotenuse: f. Opposite N: d.**
2. **0.4226, 0.9063, 0.4663**
3. **sin A = 8/17 = 0.4706, cos A = 15/17 = 0.8824, tan A = 8/15 = 0.5333**
4. **344.45 mm**
   650 × sin 32°
5. **836.41 mm**
   1,250 × cos 48°
6. **400.71 mm**
   900 × tan 24°
7. **844.62 mm**
   520 ÷ sin 38°
8. **166.29 mm**
   300 ÷ tan 61°
9. **$10\frac{7}{8}$″ (10.870″)**
   42 × sin 15°
10. **23.63°**
    $\tan^{-1}(350 \div 800)$
11. **22.02°**
    $\sin^{-1}(0.45 \div 1.2)$
12. **42.54°**
    $\cos^{-1}(700 \div 950)$
13. **62.6°**
14. **Grade 2%, angle 1.15°**
15. **14.04°**

Exercise 13B

1. **Travel 530.33 mm, run 375 mm**
   375 ÷ sin 45°, and tan 45° = 1
2. **Travel $10\frac{3}{8}$″ (10.392″), run $5\frac{3}{16}$″ (5.196″)**
   9 ÷ sin 60°, 9 ÷ tan 60°
3. **$31\frac{3}{8}$″ (31.358″)**
   12 ÷ sin 22.5°
4. **Set 320.00 mm, run 554.26 mm**
   640 × sin 30°, 640 × cos 30°
5. **30.00°**
   $\tan^{-1}(250 \div 433)$
6. **True offset 750 mm, travel 1,060.66 mm**
   √(450² + 600²) = 750, 750 ÷ sin 45°
7. **8,932 mm**
   750 ÷ tan 4.8° = 8,931.51
8. **180 mm**
   18,000 × 1 ÷ 100
9. **Rise 1,767.25 mm, floor length 8,314.25 mm**
   8,500 × sin 12°, 8,500 × cos 12°
10. **14.43 mm, a 60° groove**
    25 × tan 30°
11. **Corner 144°, cut 72°, away from square 18°**
12. **Rise 1,169.71 mm, chord 3,785.26 mm**
    3,600 × tan 18°, 3,600 ÷ cos 18°
13. **29.74°**
    $\tan^{-1}(8 \div 14)$
14. **530.25 mm, 0.08 mm short**

Exercise 13C

1. 845.05 mm
2. 4.18 m
3. $12\frac{9}{16}$″ (12.586″)
4. 25.02°
5. 38.53° and 51.47°
6. $22\frac{5}{8}$″ (22.627″)
7. 2.39°
8. 9.21 mm
9. True offset 403.11 mm, travel 465.47 mm
10. 848.53 mm. Yes, it doubled: the set and the travel keep the same ratio.

Exercise 13K

1. 14.04° and 36.87°
2. 1 in 24, 4.17%, 2.39°
3. 7.5 in 12, 1 in 1.6, 32.01°
4. 34.62°
5. 3.5 in 12 is 1 in 3.43. Asphalt shingles in normal application: no (needs 4 in 12). Low slope application: yes (needs 2 in 12). Wood shingles: yes (needs 3 in 12).
6. 2 in 12 is exactly 1 in 6, not less, so Sentence (1) applies: 150 ÷ 300 = 0.5 m².
7. Run 3,800 mm, length 3,818.95 mm, angle 5.71°
8. Rise 500 mm, angle 7.13°
9. Roof angle 22.62°, seat cut 22.62°, plumb cut 67.38°
10. 32.74°
11. Seat cut 18.43°, plumb cut 71.57° to the rafter edge

A Kitchen Math

ANSWERS

Part 1: Textbook

Try it A.1

1. 3.75
2. 0.4
3. 4.5 L
4. 2.7 kg
5. 2
6. 3,000 g

Try it A.2

1. 375 mL
2. 60 mL
3. 6 Tbsp
4. 8 cups
5. 1.361 kg
6. 227 g

Try it A.3

1. 80%
2. 80%
3. 17 kg
4. 7.5 kg
5. 13.33 kg
6. $4.00

Worksheet A1

1. **1.2 kg of trim, 80% yield**
   6 – 4.8 = 1.2 kg. 4.8 ÷ 6 × 100 = 80%
2. **4.8 kg**
   3.6 ÷ 0.75
3. **Factor 2.4: 3.6 kg of beef, 1,920 g of beans, 2.88 L of tomatoes, 72 mL of chili powder**
   24 ÷ 10 = 2.4, then multiply each amount by 2.4
4. **0.9 kg**
   6 ÷ 10 = 0.6, 1.5 × 0.6
5. **70 mL, or 14 tsp**
6. **$4.00**
   3.60 ÷ 0.90
7. **0.8333 (to four decimal places). The new batch is smaller.**
   25 × 240 = 6,000 g. 40 × 180 = 7,200 g. 6,000 ÷ 7,200

Part 2: Exercise Book

Exercise A1

1. **3.6**
2. **0.75**
3. **6.3 kg**
   42 ÷ 18 × 2.7
4. **Factor 3.5, 4.375 kg**
   70 × 200 = 14,000 g. 25 × 160 = 4,000 g.
   14,000 ÷ 4,000 = 3.5
5. **875 mL**
6. **80 mL, or 16 tsp**
7. **3.175 kg**
8. **567 g**
   20 ÷ 16 = 1.25 lb, × 0.45359237 kg
9. **1.5 kg of trim, 80% yield**
   6 ÷ 7.5 × 100
10. **8 kg**
    5.2 ÷ 0.65
11. **7.8 kg**
12. **10.345 kg, so buy 11 kg**
    6 ÷ 0.58
13. **$30.00**
    18.00 ÷ 0.60
14. **$25.00. It is less.**
    18.00 ÷ 0.72

B Practice Assessment

ANSWERS

Part 2: Exercise Book

Exercise B1

1. 17,145
2. 51,684
3. 4,137
4. 15,264
5. 149,202
6. 425,736
7. 804 R 5
8. 400 R 15
9. 332 R 24
10. 18.48
11. 0.83
12. 7.310
13. 125.090
14. $4\frac{5}{6}$
15. $5\frac{7}{8}$
16. $1\frac{1}{4}$
17. $1\frac{5}{12}$
18. $6\frac{1}{16}$
19. $\frac{9}{16}$
20. $2\frac{9}{16}$
21. $20\frac{1}{4}$
22. $3\frac{3}{4}$
23. $\frac{3}{4}$
24. 18.155
25. 257.229
26. 34.616
27. 5.372
28. 2.838
29. 38.5
30. 2.883
31. 0.625
32. 0.688
33. $\frac{8}{125}$
34. 1.35 and 0.075
35. 6.8% and 240%
36. 160% and 35%
37. $\frac{9}{20}$
38. 39
39. 87
40. 60